Supplementary Information for

Autonomous discovery of new structure-plausibility laws for explainable and rapid crystal diagnosis and screening

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Abstract

This document provides the data sources and licences, definitions of all measured quantities and laws, exact procedures used to damage structures, search and testing procedures, and the full record of eleven refuted claims. It also documents the data-split incident, reproduction instructions, additional tests of law accuracy and physical meaning, the PRIS-derived synthesis score, and applications to generated structures and database records.

Contents

S1 Autonomous discovery protocol and its audit trail

S1.1 What was autonomous

Hypotheses, feature designs, experiments, diagnostics and refutations were generated and executed by AI agents across multiple sessions. The models used across the numbered investigations are recorded in the public session archive. No detailed scientific hypothesis was supplied by a human. The agents operated in a shell with read/write access to the datasets. The programme ran under one continuous analysis protocol across sessions between 2026-07-27 and 2026-08-14, comprising 568 numbered investigations and four final analyses whose plans were recorded before evaluation (pre-registered analyses). Human input set the broad goal of finding improved crystal-structure laws and formulas, redirected the scope, and provided authorisation decisions on whether to use the reserve data that had been kept out of law development.

S1.2 How the discovery trail was audited

Unlike a conventional study, the scientific decision trail is recoverable from publicly auditable records:

Artefact	Content
Raw session archive (JSONL, 6,381 records)	timestamped user and assistant messages, tool calls and returned results. The public evidence export excludes hidden reasoning, encrypted content and system or developer records
`git log`	code state at each milestone, including the `prereg-v1.0` and `stage2-complete` tags
Script docstrings (60 files, 15,360 lines)	each script's header records *why* it was written, what it replaced, and which earlier conclusion it falsified
Workflow transcripts (14 runs)	parallel research and independent checking agents, with their individual results

The eleven refuted claims in Supplementary Fig. S1 are verifiable from preserved records rather than asserted retrospectively. Each claim has a timestamp, the code that produced it, the code that falsified it, and the document revision in which it was withdrawn.

S1.3 Scale and outcomes of the autonomous search

The statistics below cover the entire programme described above, under the same data conventions and automated checks. Source records for every session are preserved in the repository record.

Metric	Value
Wall-clock span	2026-07-27 10:31 → 2026-08-14
Investigations and final analyses	572 (568 numbered rounds + 4 pre-registered analyses on held-out data)
Candidate evaluations across the broader autonomous programme, documented lower bound	2,037,606 (scope stated below)
Analysis scripts written	615 (60 + 546 + 9) with 575 accompanying test files and 317 plans fixed before analysis
Feature tables and archived outputs	84 parquet tables (1.4 GB) + 72,583 result files (34.9 GB)
Structures analysed in the structure-law programme	99,162 experimental + 5.95 M Alexandria + 2.03 M ELEMENTA. The final analyses newly featurised 42,691 structures
PU/PSS analysis pool	The separate PU analysis pool is documented in Section S16
Tool invocations, initial session	1,095 (Bash 806, Read 110, Edit 56, Write 35), plus 14 multi-agent workflows and 41 literature queries
Conclusions stated and then self-refuted	11
Prespecified outcome and constraints for Set 4	Met: all pre-registered satisfaction and damage-detection criteria
Prespecified outcome and constraints for PSS	Met: decision-rate and structural-baseline criteria; held-out DFT comparison reported separately
Evidence retained here	8 retained laws (Law 1–Law 8) and PSS

The retained laws and the fitted score have distinct roles. Law 1–Law 8 are the physicochemical laws, and Set 4 met every pre-registered success criterion. PSS is the task-specific continuous ranking layer developed from the same structural space.

Plotting conventions of main-text Fig. 1e. One mark denotes one investigation that reached its own verdict. The four lanes contain investigations that met a success criterion, missed a success criterion, showed no gain, or were abandoned. No mark is shown for 175 investigations that produced no verdict. Of these, 131 could not run because inputs or preceding steps were unavailable, 37 had no archived verdict that could be checked, and 7 measured quantities without testing a success criterion. Investigations on adjacent screening problems are excluded from the strip. The optimal-tree analysis was evaluated only on the discovery split, so it appears as a no-gain marker rather than on the held-out curve.

Itemisation of the candidate count. The documented lower bound of 2,037,606 counts hypotheses across both the crystal-law problem and adjacent screening problems in the same programme. It sums only search sizes stated explicitly in archived round reports, where each contribution remains traceable. Search grids without a recorded size are excluded, so the true total is larger. This programme-wide count is distinct from the 8,466 candidate laws evaluated in the final Set 4 analysis. More than 99.9 per cent of documented candidates were removed before confirmation by a search plan fixed in advance or by predefined inclusion checks. Nearly all candidates that reached confirmation then failed criteria for replication, transfer to other data, or robustness across multiple settings.

S1.4 The audit catches documented errors, not unknown ones

The audit demonstrates that AI agents can conduct a multi-day empirical programme in which later checks detect errors produced within the same workflow. Each of the eleven documented failures was detected only after its claim had entered the report as a result. **We have no estimate of the number of undetected errors.** The absence of the relevant diagnostic allowed each false conclusion to survive. This omission motivated the checklist in Supplementary Fig. S1.

	Claim that did not survive	Why it looked true	Diagnostic that exposed it
R1	D2 cation swaps cannot be detected (3–14% over four searches)	Four groups of structural quantities all failed and two physical explanations seemed plausible	Madelung energy of every damaged cell was exactly zero because the damage construction was flawed
R2	Law thresholds were fitted only on the discovery split	The split file existed and was cited in the written protocol	After the analysis set grew, 3,215 rows without split labels entered one fit
R3	The satisfaction–damage-detection trade-off is a physical limit	The pattern recurred across six targets, four input groups and three analysis levels	The search used its full allowed loss of experimental-crystal satisfaction on the first law
R4	No unrestricted model can exceed 87% damage detection at 99% experimental-structure satisfaction	A boosted tree gave this result on separate parent structures	When each damage type was omitted during fitting, damage detection fell by 0.45–0.87
R5	Symmetry is the best single law (99.17% satisfaction, 39.5% damage detection)	Both numbers exceeded those of the core law	Damage detection was 0.98 for random displacement and 0.00 for expansion: a pattern of the damage procedure rather than chemistry
R6	Charge-topology quantities add nothing because removing them changes no result	The repeated analysis matched to the last digit	The removal switch deleted only unconditional laws and left the conditional versions in place
R7	Materials Project labels are noisier because the database is heterogeneous	Accuracy rose steadily with energy difference and adding label noise reproduced the trend	Three databases computed with consistent settings all scored below Materials Project in matched energy ranges
R8	Structural criteria predict energy ordering (0.6603 = 0.6603, exact)	The two values agreed to four decimal places	On the subset ranked correctly by energy the two quantities are equal by definition
R9	Density predicts synthesis history better than formation energy	0.7103 versus 0.6476, unchanged by four checks	SiO_2 supplied 43.6% of all pairs. Giving each composition equal weight reversed the result to 0.6347 versus 0.7451
R10	Pauling's rules perform worse than chance (0.14–0.36)	The values were far below chance and seemed to have a symmetry-based explanation	Ties made up 77.7% of pairs and were counted as errors. Experimentally recorded entries were actually more symmetric (space group 87 versus 62)
R11	An upper contact bound adds no value because the law must be one-sided	At 99% satisfaction the best range is [0.73, 3.04], so the upper bound detects no additional damage and damage detection (0.2823) is below the one-sided law (0.2881)	At equal satisfaction below 95%, the best upper bound changes sharply from about 3.0 to 1.08 and improves damage detection by up to 0.16

Fig. S1 Refuted-claim ledger. Each row lists a claim identifier used throughout the Supplementary Information. The three text columns record the withdrawn claim, the evidence or reasoning that initially supported it, and the diagnostic that exposed the error. Alternating grey shading separates entries, while red identifiers and green diagnostic text distinguish the audit fields.

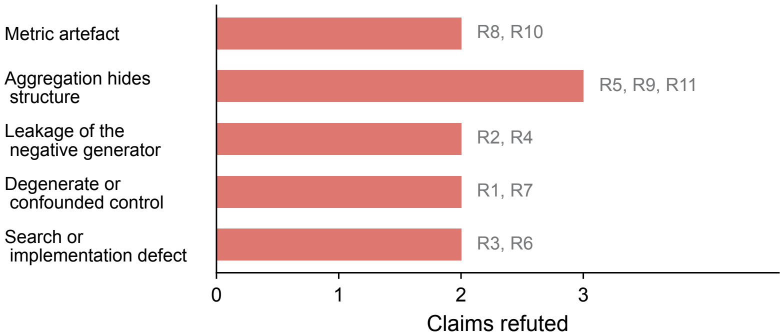

Fig. S2 Refutations grouped by failure mode. Each audited claim from Supplementary Fig. S1 is assigned once to the diagnostic category that exposed it. Categories are arranged on the vertical axis, horizontal bar length gives the number of assigned claims, and R identifiers at the bar ends link each category to the corresponding ledger entries.

S2 Data sources, licences and the analysis set

S2.1 Sources

Source	Records	Use	Licence
ICSD	73,823	experimental structures	FIZ Karlsruhe, **not redistributable**
COD	25,339	experimental structures	CC0
Materials Project	154,377	energies, ICSD linkage, experimental records	CC-BY-4.0
Alexandria PBE-3D	5,952,515	polymorph ranking, consistent settings	CC-BY-4.0
ELEMENTA (core)	2,028,008 trajectory endpoints	polymorph ranking, `max_force` available	CC-BY-NC-4.0
LeMat candidate pool	108,580	polymorph ranking	mixed
OMat24	*screened out*	—	see S2.3

S2.2 How structures entered the analysis

The analysis began with 99,162 experimental structures. Excluding unary structures, for which Pauling's rules are undefined, left 98,352. Of the full cohort, **35,490 structures (35.8%)** had both assignable formal charges and every required feature. Formal-charge assignment alone covers 64.5%, as §S2.4 explains.

S2.3 Why OMat24 was screened out

OMat24's training partition consists of AIMD snapshots at 1000 K and 3000 K and rattled configurations, which are non-equilibrium structures intended for interatomic-potential training. Energy differences there reflect thermal excursion, not phase stability. The `rattled-relax` partition does contain optimisations, but its records are **trajectory frames**. A sample of 8,001 records spanned 334 trajectories. Complete trajectories contain 24 frames, but the row-level sample ended partway through one trajectory. Retaining only the endpoints left more than one endpoint for just **2 of 332 compositions**. The partition therefore cannot form polymorph groups.

S2.4 Formal-charge assignment limits the evaluable population

No charge-balanced integer valence assignment could be found for 41.1% of 523 sampled held-out structures. The table below partitions these failures; its denominator is the subset without an assignment.

Cause	Share of failures
Mixed-valence transition metals (resolvable with non-integer mean valence)	**72.1%**
No variable-valence element: integer combinations genuinely unsolvable	27.9%

The primary Law 1–Law 8 analysis uses integer assignments. Charge-assignment coverage is the fraction of structures with assigned formal charges. In a separate analysis, a non-integer mean-valence fallback raised this coverage from 64.5% to **80.9%**. The retained laws and thresholds were applied without refitting. The four-law core was satisfied by 0.9256 of the newly admitted population, compared with 0.9577 of the integer-valence population (121 groups, 95% CI $\approx \pm 4.7$ pp).

S3 Feature definitions

We used pymatgen `CrystalNN(weighted_cn=False, x_diff_weight=0)` to identify neighbours. Valences came from composition-derived formal charges. We **never used `BVAnalyzer`**, because inferring valence from bond lengths before testing bond-valence laws is circular.

For the common-denominator comparison in main-text Fig. 2c, each Pauling criterion produced one Boolean verdict for each of 5,297 held-out structures: (1) mean rule-2 bond-strength deviation at most 0.01; (2) rule-3 edge-or-face sharing fraction at most 0.10; (3) no rule-4 high-valence violation; and (4) satisfaction of rule 5. Under the benchmark convention also used for PRIS, an unavailable measurement counts as satisfying the criterion. Experimental-structure satisfaction is the fraction satisfying the law set under this convention. The four rules were satisfied by 2,312, 598, 2,279 and 4,325 structures, respectively, and 345 satisfied all four. The separate historical audit retains each rule's natural site-orbit, orbit-pair or structure unit and is not used in Fig. 2c.

The feature tiers distinguish composition-only inputs (T0), low-cost structural inputs (T1), and quantities that require a fully specified structure (T3).

Symbol	Definition	Tier
`bl_min`	min over cation–anion bonds of d / [r(cat) + r(an)], Shannon radii by (element, oxidation state, CN)	T3
`bl_mean`	same ratio, averaged over all cation–anion bonds	T3
`mad_max`	maximum site Madelung energy from an Ewald sum with formal charges	T3
`madz_range`	range of (site Madelung energy ÷ absolute formal-charge magnitude), scale-free across chemistries	T3
`min_opp_frac`	min over ions of (fraction of its neighbours with opposite charge)	**T1**
`frac_like_bonds`	fraction of all bonds joining same-sign ions	**T1**
`econ_mean/max`	Hoppe effective coordination number, $w = \exp[1 - (d/d\text{min})^6]$	T3
`mef_mean`	Hoppe MEFIR effective-radius mismatch, signed	T3
`aa_min/mean`	ligand–ligand contact ratio, the geometric core of Pauling's first rule	T3
`ecc_max`	$\lvert\Sigma\ \hat{\mathbf{u}}_i\rvert$ / n, off-centring of the cation in its coordination shell	T3

Symbol	Definition	Tier
`p5_n_distinct`	mean number of distinct CN values per (element, oxidation state)	T1
`fi`	Pauling ionicity $1 - \exp(-0.25\ \Delta\chi^2)$	**T0**

T3 quantities require an existing structure and cannot be calculated from composition alone. They are appropriate here because this work evaluates existing structures.

S3.1 Oxidation states

For the PRIS analysis, `discriminate.guess_oxi` assigns formal oxidation states from composition alone by integer charge balancing. Native CIF oxidation-state annotations are not used. `BVAnalyzer` was never called anywhere in the pipeline, for the reason above. Structures for which no charge-balanced integer assignment can be found are retained with missing charge values. Deleting them would select against mixed-valence chemistry, which contains the most information about bond-valence laws. Missing values therefore remain missing and are handled by the convention in §S3.3 rather than being imputed.

S3.2 Fixed radius and bond-valence lookups make parameter choices reproducible

Every contact ratio and valence sum follows one fixed parameter-lookup sequence. *Shannon radii* are keyed by (element, oxidation state, coordination number). When the optional charge-assignment fallback of Section S2.4 is used, its non-integer mean valence is rounded to the nearest integer before lookup, with its sign retained. If the observed coordination number has no tabulated entry, the lookup tries CN, CN−1, CN+1 and then VI, in that order. If none is available for the element and oxidation state, it uses a representative per-element ionic radius. The final fallback is 1.0 Å for a cation and 1.4 Å for an anion. Spin-split Shannon entries are resolved through the same coordination fallback, so the radius used for a 3d ion is spin-agnostic. We did not separately quantify sensitivity to high-spin versus low-spin radii.

Bond-valence parameters come from the IUCr `bvparm2020` compilation. They are keyed by cation element and integer valence, and by anion element and integer valence. For each pair, the lookup takes the first-listed parameter set and its tabulated b, never a generic constant. Pairs without tabulated parameters are skipped and counted. Every bond-valence statistic reports the fraction of neighbour entries with available parameters (`bv_param_cov`). A structure-level Law 8 value is emitted only when the structure contains at least two parameterised cation–anion neighbour entries. Its mean deviation includes all non-zero-charge sites and is normalised by the formal charge.

S3.3 Neighbour-algorithm robustness

CrystalNN is the primary neighbour algorithm. For robustness, we recomputed BrunnerNN and MinimumDistanceNN on a common random subsample of 3,000 structures (`random_state=7`). The comparison quantifies the sensitivity of Law 1 and the individual Pauling criteria. The exact spreads, in percentage points, were 0.18 for $\rho \geq 0.735$, 5.74 for Pauling's fourth rule, 4.05 for the first, 3.93 for the third, 1.81 for the fifth and 0.83 for the second. Thus the contact-ratio criterion was 32-fold less sensitive than the most fragile criterion within this algorithm family. Because all three algorithms share pymatgen's tolerance conventions, the 0.18-percentage-point spread brackets disagreement within that family of definitions.

S3.4 Missing values and minimum feature availability

A missing feature value counts as satisfying a law for benchmark calculations. Only features computable on more than 0.9 of experimental structures are therefore admitted to the search. This prevents a criterion from receiving high satisfaction merely because values are unavailable. When applied to a new structure, a missing charge or required feature produces no verdict and routes the structure to review.

S4 Searching for simple crystal-plausibility laws

S4.1 The search balances satisfaction against damage detection

Here, satisfaction is the fraction of experimental structures that satisfy a law set under the missing-value convention above. Damage detection is the fraction of damaged structures that fail it.

The candidate model is a set of N single-feature threshold laws joined by AND. Its objective is to maximise damage detection on composition-matched damaged structures. Experimental-structure satisfaction must remain above a stated minimum. An optional second constraint requires every chemical group's satisfaction to exceed the same stated group minimum.

S4.2 Executable checks exposed two search defects

(a) The greedy step used the entire allowed loss in satisfaction. Beam search ranked candidates only by damage detection. At depth 1, it therefore selected a law whose satisfaction was exactly at the required minimum, leaving no room for another law. At a required satisfaction of 0.99, this produced a single-law set. The complete repair allocates half the beam to candidates ranked by *damage detection gained per unit reduction in satisfaction.* The other half remains ranked by total damage detection plus the weakest class's rate. Damage detection at a required satisfaction of 0.95 rose from 0.658 to 0.734.

(b) Ban-filter failure and correction. Conditional laws use the composite key `"G:<guard>:<target>"`. Filtering only on the plain column name therefore removed unconditional variants but left conditional variants in the search. Several feature-removal tests reported as "identical after removal" although they had removed nothing. The defect is recorded as R6 in Supplementary Fig. S1, and the corrected ablation is summarised in Section S7.

S4.3 A fixed search grid limits how much the search can adapt to the data

A candidate law is a (feature, direction, percentile-threshold) triple. Each law set can contain at most one threshold per feature column. Conditional laws take the form "if G then T" and are evaluated as $\neg G \vee T$. The guard G is restricted to T0 and T1 selectors, so it never requires the metric structure that the law judges. Beam search maximises damage detection subject to minimum overall satisfaction and, optionally, minimum satisfaction for each anion group.

All values in this subsection are from the discovery split. At a required satisfaction of 0.99, the search returned five laws with satisfaction 0.9917 and damage detection 0.304. At 0.98, it returned three laws with satisfaction 0.9813 and damage detection 0.4161. At the recommended value of 0.95, it returned five laws with satisfaction 0.9539, damage detection 0.6224, and weakest-group satisfaction 0.928. This operating-point sequence uses the repaired allocation described above, not a further refinement. Beam width, minimum gain per added law, and the percentile grid are fixed in the search implementation.

S4.4 Composition controls prevent misleading cross-database comparisons

Structure counts, accessible energy spans and the concentration of pairs in one composition differ strongly among ranking databases. We therefore report the source populations and repeat the comparison inside a common hull-energy window (Supplementary Fig. S3a,b). Every accuracy is first computed within a composition and then averaged with equal weight across compositions. This prevents a database with many near-duplicate pairs from dominating the comparison and separates dataset composition from the structural signal being tested.

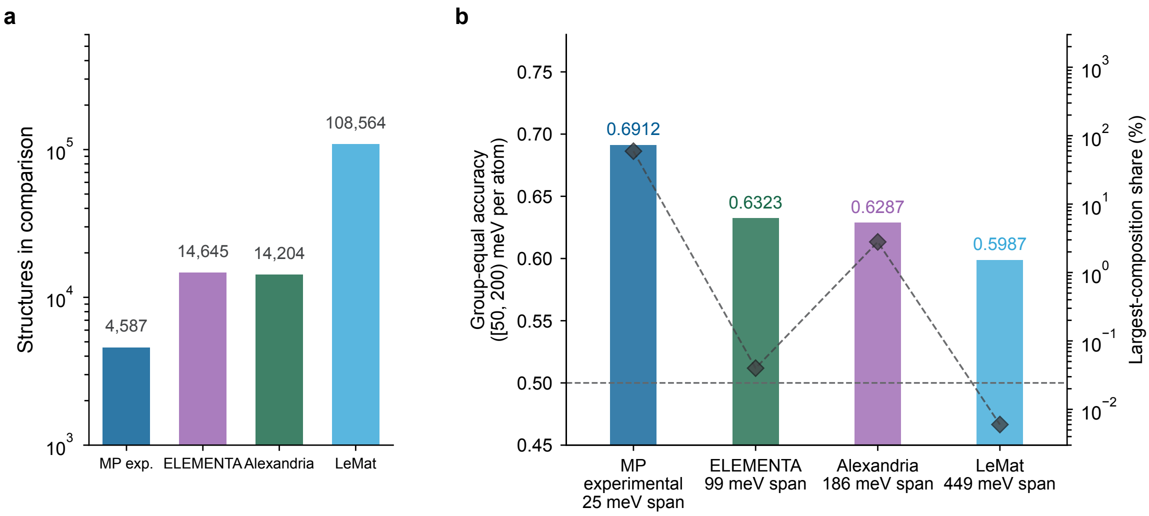

Fig. S3 Ranking-data scale, energy-span matching and composition concentration. **a**, Database-specific structure counts on a logarithmic vertical axis. Labels above the bars give the counts, and colours identify the source databases. **b**, Composition-group-equal ranking accuracy within the common hull-energy window. Bars report accuracy on the left axis. Diamonds and a dashed line report the largest composition's pair share on the logarithmic right axis. Horizontal-axis sublabels give the intra-composition energy spans. The horizontal dashed line marks chance accuracy.

S5 Composition-preserving perturbations isolate structural failure modes

Five damage types, each preserving composition, stoichiometry and atom count, were retained. Here, ΔE_{M} is the change in Madelung energy relative to the experimental parent:

Class	Operation	Median ΔE_M (eV per atom)	Median `bl_min`	Verdict
—	(experimental structures)	—	0.937	—
D1	uniaxial compression 15–30%	$\mathbf{-2.100}$	0.791	retained, flagged
D2	cation–cation swap (formal charges must differ)	+0.929	0.814	valid
D3	random displacement 0.3–0.8 Å	+0.090	0.521	valid
D4	isotropic expansion 20–40%	+6.278	1.234	valid
D5	cation–anion swap	**+5.803**	**0.930**	valid despite being geometrically invisible
~~S6~~	~~shear strain 10–25%~~	−0.004	0.881	**discarded**

D1 is flagged because its Madelung energy *decreases*: compression shortens cation–anion distances and increases Coulomb attraction. Pauling's electrostatic framework contains no short-range repulsion. The supplementary `bl_min` measure supplies this missing contact term and identifies the compressed structures as unreasonable.

S6 was discarded because neither independent probe confirmed it as unfavourable. Its median ΔE_M was −0.004 eV per atom, and only 10% of cases fell below the `bl_min` threshold. These independent energetic and geometric probes therefore did not support retaining S6. Doing so would have repeated the unverified-damage design failure exposed by R1, the D2 cation-swap implementation error described in Section S12.

D5 is the informative damage type. Geometrically, it is nearly indistinguishable from experimental structures (`bl_min` 0.930 versus 0.937). Electrostatistically, it is catastrophic. No bond-length criterion detects it well (damage detection 0.03–0.21). The audited Law 6 like-charge ban closes this blind spot. Adding Law 6 to the four-law core raises held-out D5 damage detection from 0.262 to 0.576, a gain of 0.314. The largest gain on any other class is 0.057. The complete damage-type comparison on the discovery and held-out splits is shown in Supplementary Fig. S4.

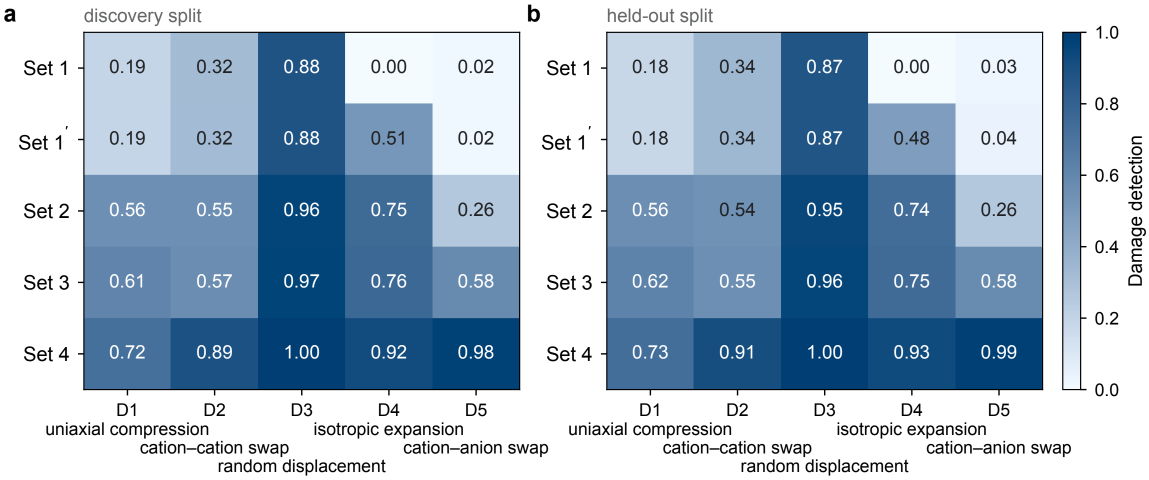

Fig. S4 Damage detection by perturbation class on the discovery and held-out splits. **a**, Discovery split used for law selection. **b**, Held-out split evaluated with the law sets fixed. Rows are PRIS law sets and columns are the D1–D5 composition-preserving perturbations. Cell colour and printed text encode the within-class damage-detection fraction on a shared scale, and the column labels define the perturbation operations.

S6 The eleven refuted conclusions

Supplementary Fig. S1 provides an overview of all eleven refutations, including why each claim was initially persuasive and which diagnostic exposed it. Supplementary Fig. S2 groups the same events by the diagnostic failure mode that exposed them. The accompanying machine-readable ledger preserves the complete trace for every event. Two cases deserve extended treatment below.

S6.1 R8: an identity mistaken for evidence

We observed that a structural score's accuracy on the energy-correct subset equalled its sign agreement with energy to four decimal places ($0.6603 = 0.6603$). We initially interpreted this identity as evidence that the score predicted energy ordering.

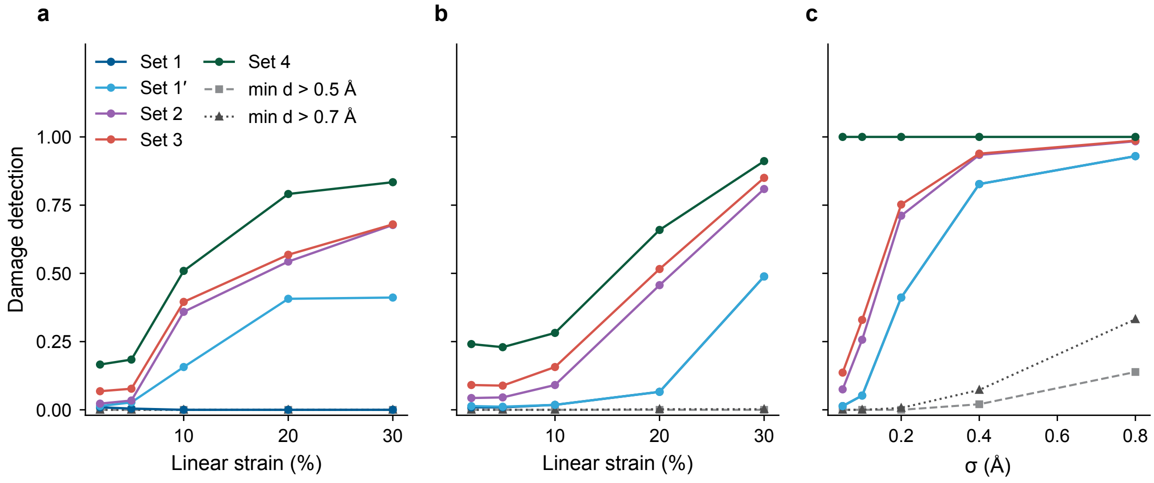

Fig. S5 Damage detection versus perturbation amplitude. All panels use the same fixed parent-structure cohort, with the vertical axis giving the fraction of perturbed structures detected. **a**, Isotropic expansion. **b**, Uniaxial compression. **c**, Gaussian displacement. Solid lines with circles identify PRIS law sets, while dashed lines with squares and dotted lines with triangles identify the two minimum-distance criteria.

On that subset, $y = \mathbb{1}[-\Delta\mathrm{E} > 0]$ by construction, so

accuracy $= \mathrm{mean}(\mathbb{1}[\mathrm{s}{>}0] = y) = \mathrm{mean}(\mathbb{1}[\mathrm{s}{>}0] = \mathbb{1}[-\Delta\mathrm{E}{>}0]) =$ sign-agreement.

Recomputing across eight criteria gave differences of exactly +0.0000 in all eight cases. **A four-decimal agreement that holds for every criterion is a definition, not a finding.**

S6.2 R10: a lesson recorded, then repeated

Supplementary Section S13 requires tie and error rates to be reported separately. Pauling's fifth rule illustrates why: it yields a non-tied comparison for only 0.2230 of pairs and ties in 88.4% of composition groups. Its accuracy therefore changes substantially when ties are credited, excluded, or counted as errors. The later synthesis-target analysis nevertheless counted 77.7% ties as errors. This scoring error produced the claim that Pauling's rules perform worse than chance (0.14–0.36). The proposed symmetry mechanism was also refuted: experimentally recorded entries are *more* symmetric, with medians of 87 versus 62.

Recording a lesson does not ensure that it is applied. The corrected diagnostics are now automated checks in the pipeline, not only statements in the document.

S7 Which feature families improve damage detection

The table reports each feature family's marginal contribution to law-set damage detection at minimum satisfaction 0.95 on five discovery-split damage types.

Family added	Damage detection	Increment
Coordination statistics + Pauling quantities (baseline)	0.509	—
+ Shannon-radius bond ratios	**0.734** (four types)	**+22.5 pp**
+ electrostatics / Madelung	0.6224 (five types)	`mad_max` enters, forced by D5
+ charge topology	0.6224	**0** (two independent lines of evidence)
+ angular (`ecc`, `angvar`)	0.6254	**0**
+ symmetry	—	**excluded**, see R5

The values 0.734 and 0.6224 are not directly comparable because the former covers four damage types whereas the latter includes D5. Shannon-radius bond ratios provide the largest gain on the first four types, while adding D5 brings the electrostatic `mad_max` feature into the selected set.

S8 Split discipline, revisions and the reserve split-labelling error

The pre-registration is preserved at git tag `prereg-v1.0` and commit `ec4b464`. Seed 20260728 assigned structures to three partitions: discovery (22,925), held-out (9,634), and sealed reserve (5,748). Assignment used a seeded hash of each structure identifier rather than composition. Structures sharing a composition can therefore cross partitions. The resulting discovery-to-held-out changes in satisfaction and damage detection are shown overall in Supplementary Fig. S6 and by damage type in Supplementary Fig. S4.

Revision R1 corrected the design-effect measurement. **Revision R2** corrected the recorded source of oxidation-state assignments. Both were committed as `da563b1` **after** the analysis plan was fixed, and their timestamps are disclosed here.

The reserve split-labelling error. After the plan was fixed, the analysis set was widened to all experimental structures. Newly admitted rows had no split label. A full-sample percentile-threshold fit consequently included 3,215 reserve rows in the law analysis and 863 in the historical formula fit. An `assert_clean()` check now refuses to fit if any row lacks a split label or belongs to the reserve. **Because reserve rows influenced this threshold fit, the reserve can no longer serve as a fully independent final test.**

The first authorised evaluation of the reserve. Explicit authorisation opened the reserve on 2026-08-01 (opening index 1), using one of three permitted evaluations. The analysis tested V4, a candidate five-law refinement that adds an integer-valence condition to the like-charge ban. Its success criteria were fixed before evaluation. Overall damage detection had to increase by $+0.02$, or detection in the weakest damage type by $+0.03$, while satisfaction could change by no less than -0.005.

The reserve contained 3,163 evaluable experimental structures and 2,195 damaged structures. V4 changed satisfaction by -0.0003, overall damage detection by $+0.0105$, and detection in the weakest baseline class, D1 uniaxial compression, by -0.0102. Thus satisfaction remained within its limit, but neither performance alternative met its gate. D5 detection nevertheless increased by $+0.182$ as an additional class-specific result. V4 is therefore refuted at its stated effect size and does not appear in the main text.

Two findings extend beyond V4. First, the integer-valence condition behaved as intended on reserve data. Without it, nitride-group satisfaction fell by 5.8 percentage points; the condition recovered all but 0.7 percentage points. This confirms fractional-valence polyanions as the natural exception to a like-charge-contact law. Second, the effect size was $+0.0238$ on the repeatedly reused held-out split and $+0.0355$ on post-2019 data, but only $+0.0105$ on the reserve. This is the project's direct measurement of effect inflation after repeated held-out use. It also shows that a time-separated dataset cannot replace an untouched reserve. Two permitted evaluations remain.

Composition-unseen sensitivity. Because the structure-identifier hash permits composition overlap, we performed an additional outcome-blind subgroup analysis using only the discovery and held-out partitions. Reduced formulas defined composition groups, and each chemically damaged structure inherited the reduced composition of its experimental parent. The PRIS laws, thresholds and missing-feature convention remained frozen. Of 5,297 held-out experimental structures, 3,631 (68.55%) shared a reduced composition with discovery and 1,666 (31.45%) did not; the corresponding counts among 3,612 damaged structures were 2,406 and 1,206. Confidence intervals used 10,000 percentile-bootstrap resamples of whole reduced-composition clusters (seed 20260829).

For Set 4, experimental-structure satisfaction was 81.38% (95% interval, 79.73–82.99%) for composition-shared structures and 82.71% (80.65–84.67%) for composition-unseen structures; damage detection was 90.73% (89.53–91.92%) and 91.87% (90.28–93.42%), respectively (Supplementary

Fig. S6 PRIS law sets on the discovery and held-out splits. a, Fraction of experimental structures satisfying each law set. **b**, Fraction of damaged structures detected, pooled across perturbation classes. Pale bars show the discovery partition used to select the thresholds, and solid bars show held-out evaluation without refitting. Labels above paired bars report the held-out value minus the discovery value, and both panels use the same fractional scale.

Fig. S7). Weighting reduced compositions equally gave 82.04% satisfaction and 91.64% detection in the unseen subgroup. A stricter subset whose exact element sets were absent from discovery retained 83.17% satisfaction among 1,046 experimental structures and 91.49% detection among 799 damaged structures. Bond-valence coverage was lower in the composition-unseen subgroup (83.25% for experimental and 81.01% for damaged structures) than in the shared subgroup (85.90% and 89.24%); counting an unavailable value as satisfying a law can raise satisfaction but cannot inflate damage detection. This analysis is a sensitivity test within the existing held-out partition, not a newly collected external holdout; "chemical system" denotes the exact element set rather than a broader chemical family.

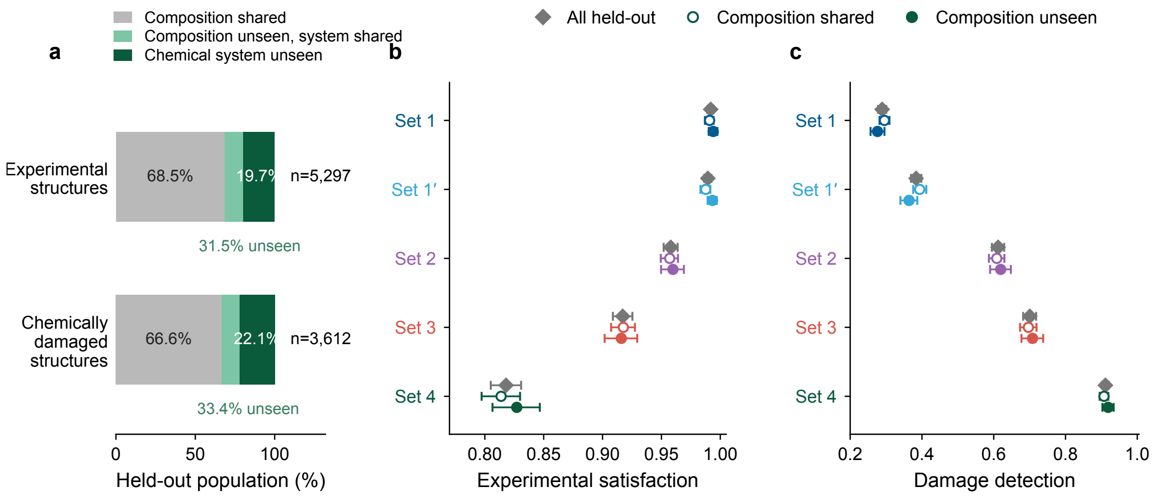

Fig. S7 Frozen PRIS law sets retain their performance on compositions unseen during law discovery. **a**, Chemistry overlap created by the original structure-identifier split. Held-out experimental structures and their chemically damaged counterparts are partitioned into compositions present in discovery (grey), unseen compositions whose element set was seen (light green), and unseen exact chemical systems (dark green). Damaged structures inherit the composition of their experimental parent. **b**, Experimental-structure satisfaction for the five frozen PRIS law sets in the full held-out population (grey diamonds), composition-shared subset (open circles) and composition-unseen subset (filled circles). **c**, Damage detection for the same populations. Points are structure-weighted estimates; bars are 95% percentile intervals from 10,000 bootstrap resamples of whole reduced-composition clusters (seed 20260829). The rules, thresholds and missing-feature convention were not changed.

S9 Reproducing the analysis

```
# example command-line configurations

# archived five-law set: Law~1 and Law~3--Law~6

python src/apply_rules.py <structure files>

# archived four-law set: Law~1 and Law~3--Law~5

python src/apply_rules.py --set core4 *.cif

# single reduced-distance law, threshold 0.735

python src/apply_rules.py --set single *.cif


# per-law values plus archived exploratory formula

# (not its historical fitted accuracy)

python src/apply_rules.py --verbose --formula a.cif


# figures read paper/data/*.csv

# plotting code does no computation
```

```
python src/paper_figs.py


# rebuilding aggregate CSVs first requires the external
# feature store
PRIS_FEATURES=/path/to/features \
  python src/paper_data.py
```

Public benchmark. Because ICSD is not redistributable and ELEMENTA is CC-BY-NC-4.0, the benchmark released with this work is built exclusively on the 25,339 COD structures (CC0).

S10 Exact search does not remove omitted-class sensitivity within the tested shallow-tree space

Purpose. To test whether the beam search of §S4 missed a better law set, we used an exact algorithm that proves optimality within a defined class of trees.

Set-up. The discovery matrix contained 12,632 experimental structures, 8,590 damaged structures, and 79 numerical features available for more than 0.9 of rows. We supplied this matrix to `pystreed` 1.3.9, which implements STreeD separable-objective optimal decision trees by dynamic programming. We set `continuous_binarize_strategy="quantile"` and `n_thresholds=3`. These settings select candidate splits from each feature's percentiles. The class of depth-≤ 3 trees strictly contains the class of conjunctions explored by beam search. Wall-clock time was 0.8 s at depth 1, 0.7 s at depth 2, 6–28 s at depth 3, and 549 s at depth 4, which was impractical.

Moving along the trade-off curve. Changing the optimisation objective did *not* move the curve: `f1-score` and `accuracy` returned the same depth-3 tree. The **cost ratio** did move the curve. It balances the cost of rejecting an experimental structure against accepting a damaged structure, and was implemented by sample replication:

cost ratio	satisfaction	damage detection
1.00	0.8895	0.8553
1.50	0.9265	0.7849
1.75	0.9367	0.7540
2.00	0.9367	0.7540
2.25	0.9367	0.7540

cost ratio	satisfaction	damage detection
2.50	0.9518	0.6732
3.00	0.9807	0.5637
4.00	0.9790	0.5697

Testing on a damage type omitted during fitting. We refitted at cost ratio 2.5 with one damage type removed entirely, then tested on that type:

damage type omitted	damage detection on types used in fitting	damage detection on omitted type
D1 uniaxial compression	0.7553	0.2222
D2 cation–cation swap	0.7048	0.1445
D3 random displacement	0.5310	0.6971
D4 isotropic expansion	0.6903	0.0235
D5 cation–anion swap	0.6747	0.0715
mean	**0.6712**	**0.2318**

The single-threshold criterion, selected by the same procedure, achieved 0.2819 on seen damage types and 0.2771 on omitted types. For each method, we first took the absolute seen-versus-omitted detection difference in each of the five leave-one-class-out fits and then averaged those five differences. The result was 0.506 for the tree and 0.321 for the threshold, a factor of 1.58. For the threshold, the near-zero signed difference between its seen and omitted means is misleading because class-specific improvements such as D3 can cancel losses such as D4 and D5 (Supplementary Fig. S13a).

First, the provably optimal depth-3 tree at satisfaction 0.95 did not select ρ. It used `ecc_mean`, `mad_an_mean`, `mad_range`, `frac_like_bonds`, `pair_per_cat`, and `cat_an_ratio`.

Second, the direction and magnitude of the change were damage-type specific. This pattern is consistent with sensitivity to the perturbation procedure, although it does not isolate the cause.

Independent replication. These numbers were produced twice, by separate agents in separate environments, from the same stored feature matrix. All values agree to four decimal places.

What this result supports. Within the tested discretized class of depth-≤ 3 trees, the in-distribution gain does not survive on average when one damage type is omitted during fitting. This establishes sensitivity to how damaged structures were generated within this feature grid. It does not show that the perturbation procedure is the only limitation or rule out gains from other features, model classes or objectives.

S11 What a one-sided criterion misses, and the condition that repairs it

What it misses. ρ is a lower bound on the reduced contact ratio, so it cannot detect a structure that is too open. Damage detection against isotropic expansion of 20–40% is 0.002 for $\rho \geq 0.735$ and 0.002 for $\rho \geq 0.804$. The complete threshold sweep shows this expansion blind spot across the tested range (Supplementary Fig. S8).

An unconditional upper bound adds no value when required satisfaction is at least about 95%. We scanned τ_lo $\in$ [0.60, 1.00] $\times$ τ_hi $\in$ [1.00, 3.60]. One-sided and two-sided criteria were compared at the same required satisfaction:

minimum satisfaction	best one-sided	best two-sided	gain
0.99	$\rho \geq 0.735$, 0.2881	[0.73, 3.04], 0.2823	−0.006
0.98	$\rho \geq 0.780$, 0.3674	[0.78, 3.26], 0.3682	+0.001
0.97	$\rho \geq 0.800$, 0.4064	[0.80, 2.80], 0.4097	+0.003
0.96	$\rho \geq 0.815$, 0.4377	[0.81, 2.40], 0.4343	−0.003
0.95	$\rho \geq 0.830$, 0.4690	[0.83, 3.58], 0.4695	+0.000
0.94	$\rho \geq 0.835$, 0.4785	**[0.75, 1.12], 0.5126**	**+0.034**
0.92	$\rho \geq 0.850$, 0.5073	**[0.80, 1.08], 0.6179**	**+0.111**
0.90	$\rho \geq 0.860$, 0.5227	**[0.83, 1.08], 0.6806**	**+0.158**

The optimal upper bound changes abruptly with the allowed loss in satisfaction. It is effectively infinite above 0.95 and snaps to ≈1.08 below it. Our initial report included only the 0.99 row, producing the false conclusion logged as R11 (Supplementary Fig. S9).

Chemical character of the upper tail. The 626 experimental structures with $\rho > 1.08$ (4.97% of 12,601) lie outside the ionic domain:

quantity	median, $\rho \leq 1.08$	median, $\rho > 1.08$
Pauling ionicity f_i	0.508	0.111
electronegativity difference $\Delta\chi$	1.683	0.688

quantity	median, $\rho \leq 1.08$	median, $\rho > 1.08$
global instability index	0.211	1.171
MEFIR mismatch (max)	0.008	1.956
Pauling rule 2 satisfied (frac. of sites)	0.833	0.000
Shannon packing fraction	0.659	0.093

The Shannon ionic-radius model does not apply to these structures, even though they satisfy the charge-balance filter. Charge balance alone therefore does not establish applicability of the ionic-radius model. Supplementary Fig. S11 resolves the response by anion chemistry and shows where the ionicity condition adds damage detection.

The conditional law. Above $\rho = 1.05$ lie 0.62% of ionic experimental structures (f_i > 0.5), 10.07% of non-ionic experimental structures, and 96.14% of expanded structures. These distributions motivate the following conditional law:

Set 1′ $\rho \geq 0.735$ **and** (f_i ≤ 0.50 **or** $\rho \leq 1.05$)

	satisfaction	damage detection	D1	D2	D3	D4	D5
Set 1 (one-sided)	0.9900	0.2881	0.192	0.324	0.879	0.002	0.019
two-sided, unconditional	0.9900	0.2823	0.173	0.307	0.875	0.011	0.018
Set 1′ (conditional)	**0.9870**	**0.3879**	0.193	0.324	0.880	**0.513**	0.023

The full law-set sequence progressively closes the expansion blind spot (Supplementary Fig. S10).

Testing Set 1′ on a damage type omitted during fitting. The condition and threshold were selected using only the four retained classes, then applied to the omitted class:

damage type omitted	threshold selected	damage detection, omitted type	one-sided baseline	gain
D1	1.05	0.1954	0.1949	+0.0005
D2	1.05	0.3243	0.3235	+0.0008

damage type omitted	threshold selected	damage detection, omitted type	one-sided baseline	gain
D3	1.05	0.8798	0.8793	+0.0005
D4	**1.05**	**0.5127**	**0.0024**	**+0.5102**
D5	1.05	0.0226	0.0189	+0.0037

The same threshold was selected in all five repeats. Its performance on each omitted type isolates the contribution of the conditional upper-contact law (Supplementary Fig. S13b).

Stated limitation. Omitting one damage type during fitting tests whether the selected threshold works on that type. This leave-one-type test does not establish whether the mathematical form would have been discovered without prior knowledge of that type of damage. We tested a two-sided band because the controlled-damage benchmark included isotropically expanded structures. A different damage type may require reconsidering the form, not only the threshold.

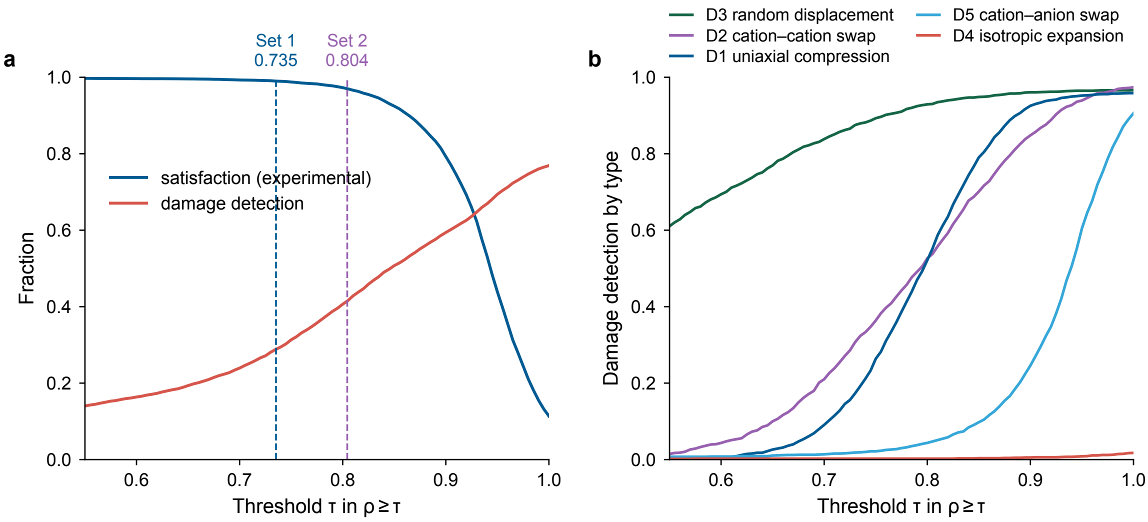

Fig. S8 Sensitivity to the Law 1 contact threshold. **a**, Experimental-structure satisfaction and pooled damage detection across a discovery-split sweep of the lower-contact threshold τ in $\rho \geq \tau$. Dashed vertical lines identify thresholds used by the labelled PRIS law sets. **b**, The same sweep resolved by the five composition-preserving perturbation classes. Colour identifies the class.

S12 Exact perturbation operators used in D1–D5

The exact procedures are reproduced below:

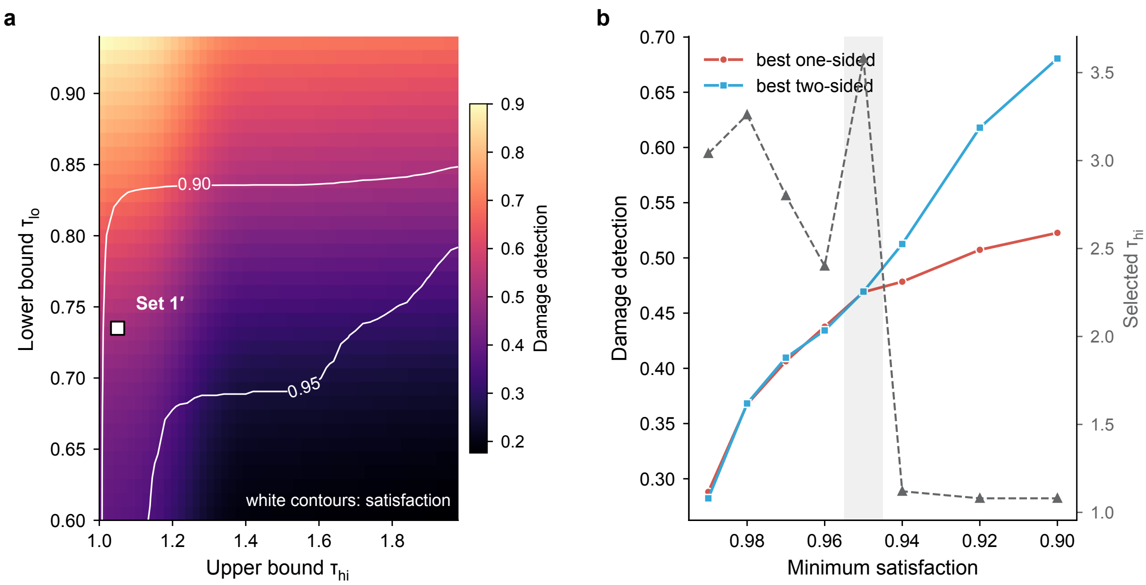

Fig. S9 Two-dimensional scan of contact-band bounds. a, On the discovery split, lower and upper reduced-contact bounds define the axes. Colour maps pooled damage detection. White contours mark experimental-structure satisfaction, and the square marks Set 1′. **b**, Minimum satisfaction is plotted against the best one-sided and two-sided detection rates. Grey triangles on the right axis give the selected upper bound.

class	operation
D1	anisotropic compression: one randomly chosen lattice axis scaled by $1 - u$, $u \sim \mathrm{U}(0.15, 0.30)$
D2	cation–cation swap between two cations whose **formal charges differ by $\geq$ 1**, with the charges swapped alongside the species
D3	random displacement of every site, Gaussian, amplitude $u \sim \mathrm{U}(0.3, 0.8)$ Å
D4	isotropic expansion: all lattice vectors scaled by $1 + u$, $u \sim \mathrm{U}(0.20, 0.40)$
D5	cation–anion swap: composition and stoichiometry unchanged, charge arrangement destroyed

The charge-difference requirement in D2 was the subject of refutation R1. The original implementation swapped species without swapping charges. Consequently, the Madelung energy change was exactly zero for 100% of samples. The resulting “structure” was only a relabelling of the original.

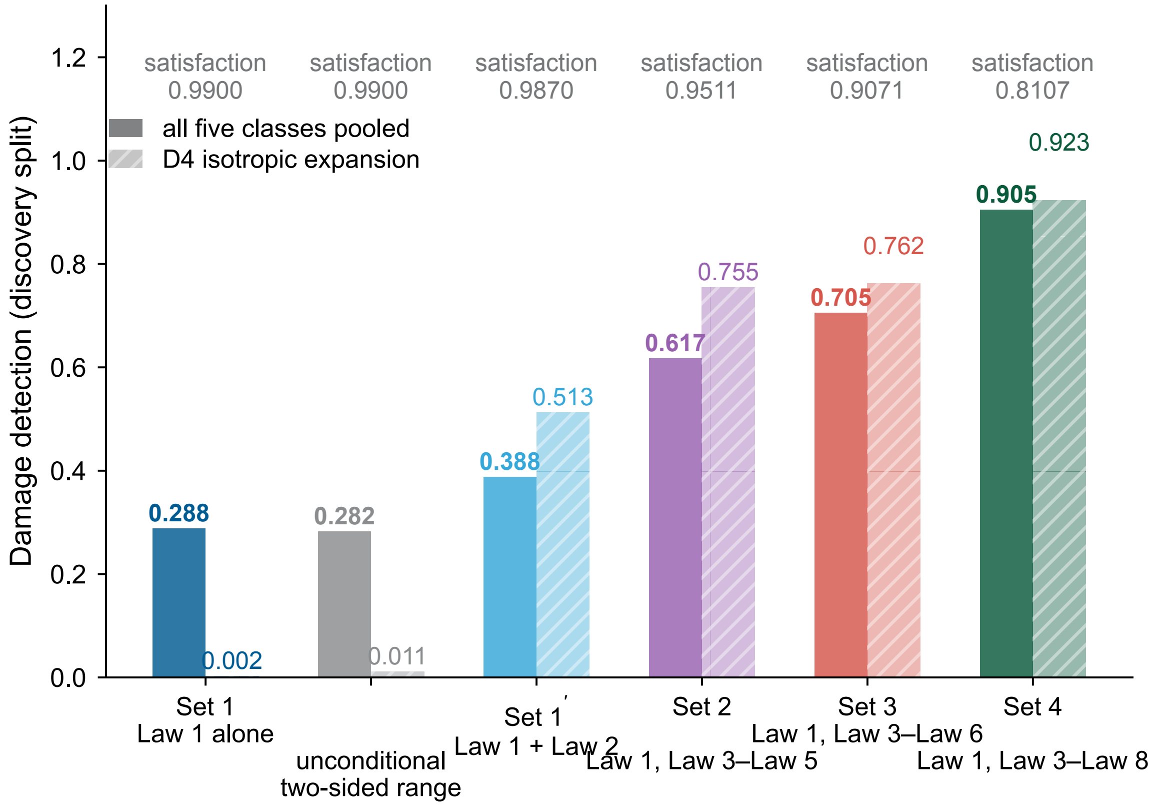

Fig. S10 PRIS law sets and an unconditional two-sided contact band. Discovery-split criteria are arranged from Law 1 through the successive PRIS law sets. The unconditional two-sided range is included as a comparator. Solid bars show pooled detection across perturbation classes, whereas hatched bars isolate isotropic expansion. Colours identify the criteria. Text above each group gives experimental-structure satisfaction.

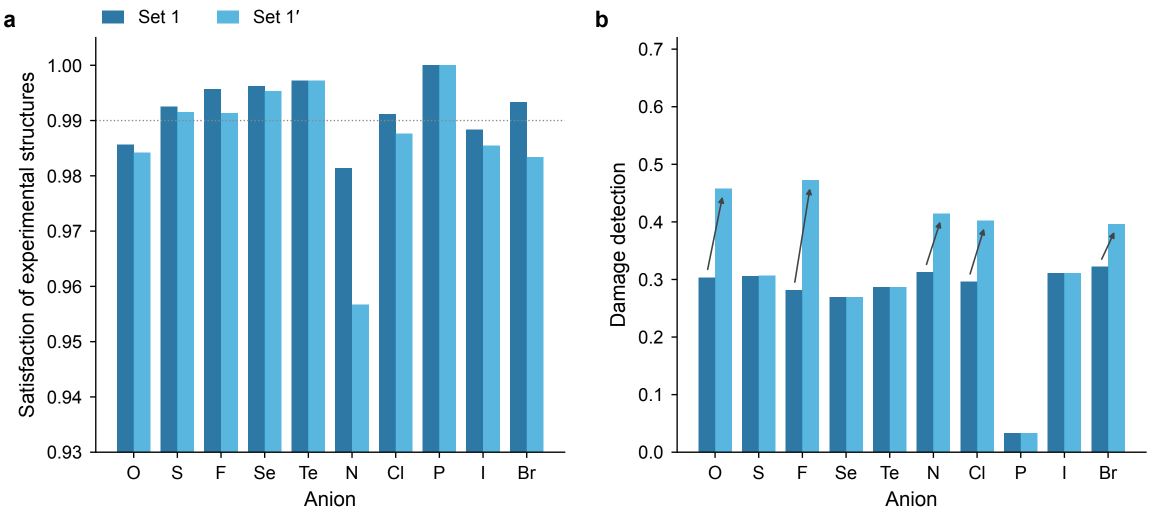

Fig. S11 Contact-law performance by dominant anion family. Discovery-split structures are grouped by dominant anion along both horizontal axes. **a**, Paired bars compare experimental-structure satisfaction for Set 1 and conditional Set 1′. The dotted line marks the reference level. **b**, Paired bars show pooled damage detection for the same criteria and families. Arrows mark differences selected by the display rule; they do not denote statistical tests.

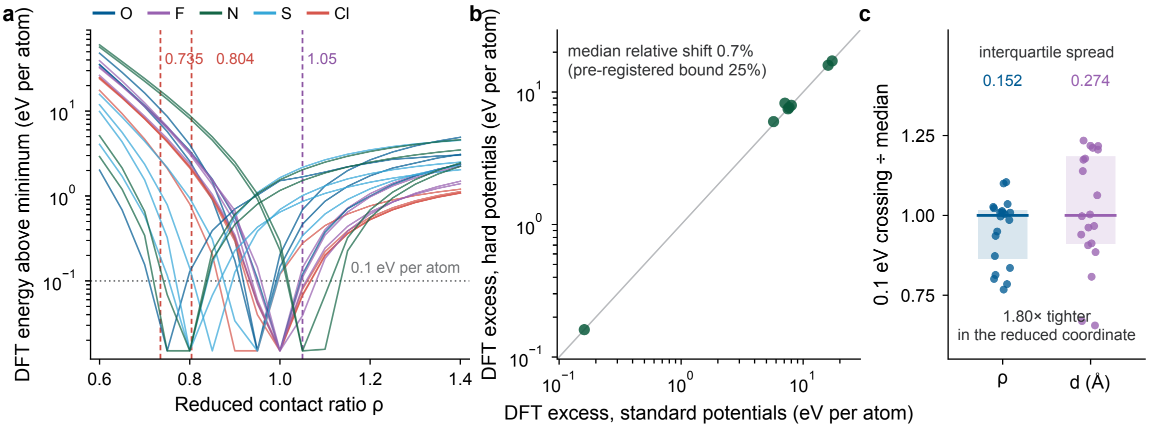

Fig. S12 The contact thresholds measured from first principles. **a**, Each line is one of twenty experimental compounds rigidly scaled along the reduced-contact coordinate and evaluated with plane-wave DFT. The vertical axis is the DFT energy above each compound's own minimum, and colour gives the dominant anion. The dotted horizontal line marks 0.1 eV per atom, the pre-registered cost of a chemical rather than geometric short contact. Dashed vertical lines mark the Law 1 floors of 0.735 and 0.804 and the Law 2 ceiling of 1.05. Every chemistry rises steeply below the floors, so the thresholds do not average soft and stiff cases. **b**, Eight compounds were recomputed with hard potentials, which freeze smaller cores, and compared with the standard set. Both axes report DFT energy excess at the floor, and the diagonal marks equality. The annotation gives the median relative shift against the threshold fixed before calculation. **c**, Each point is the contact at which a compound first pays the pre-registered cost. Reduced-contact and Å crossings are each normalised by their own median. Boxes span the interquartile range, and the number above each box gives its width. The crossing is 1.80 times more tightly localised in the reduced coordinate, enabling comparison across chemistries.

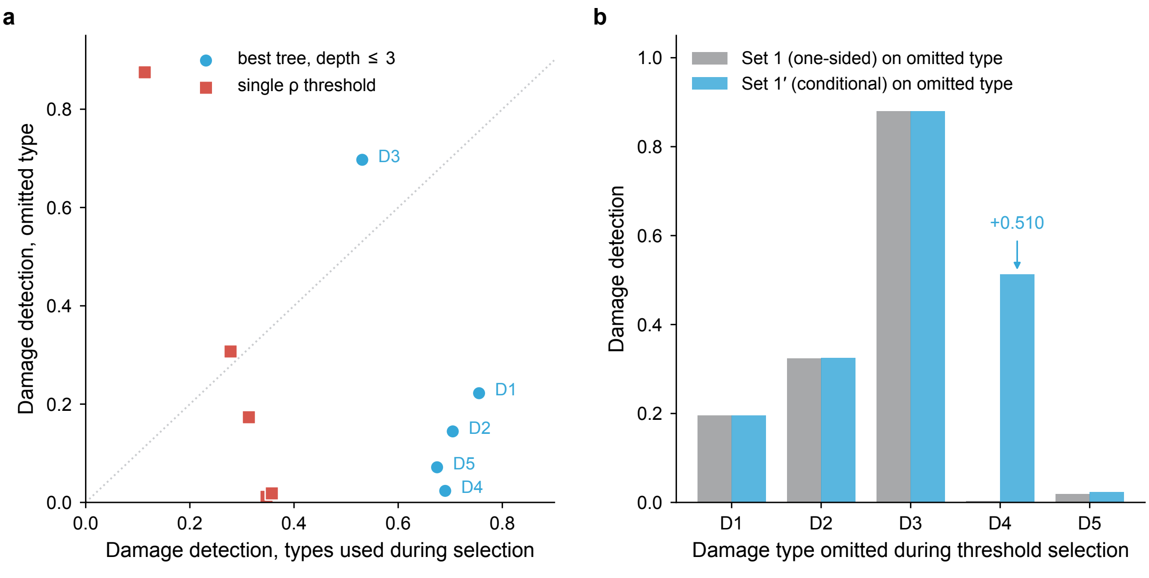

Fig. S13 Leave-one-perturbation-out evaluation of selected contact criteria. **a**, Each point represents one perturbation class omitted during selection. The axes give damage detection on the retained classes and on the omitted class. Purple circles denote the depth-limited tree, red squares denote the single ρ threshold, and the dotted diagonal marks equality. **b**, Paired bars compare one-sided Set 1 and conditional Set 1′ on each omitted class.

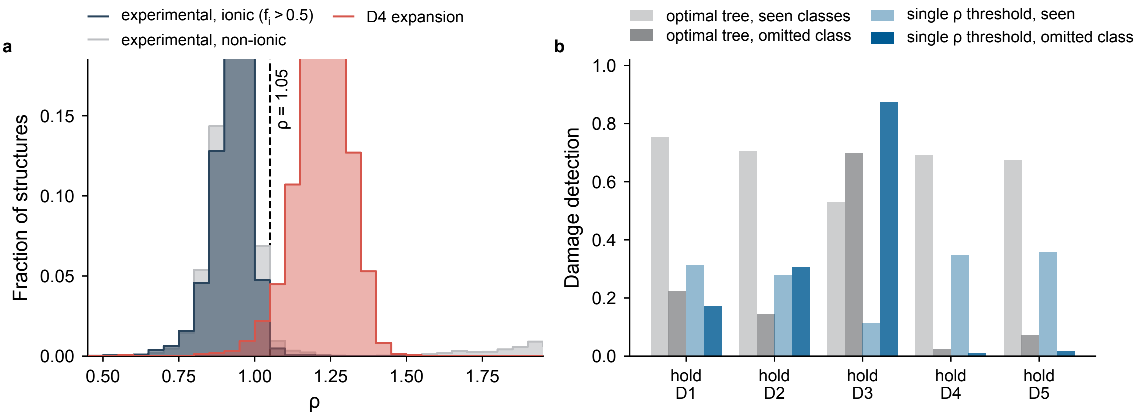

Fig. S14 Conditional contact thresholds and leave-one-damage-class-out evaluation. a, Normalised reduced-contact distributions for experimental structures separated by ionicity and for isotropically expanded structures. Filled step curves identify the cohorts, and the vertical dashed line marks the conditional upper bound. **b**, Damage detection for a depth-limited tree and a single ρ threshold when each perturbation class is omitted in turn. Grey and blue identify the two model classes, while pale and solid bars distinguish retained and omitted perturbation classes.

S13 Composition-grouped ranking separates decision rate from accuracy

Within each composition, we paired every experimentally recorded Materials Project entry with every entry lacking an experimental record. We restricted the analysis to charge-balanced ionic compositions. The resulting cohort contained 1,508 compositions, 6,758 structures, and 18,920 pairs.

A criterion provides a usable pairwise comparison when its two values differ beyond numerical tolerance. A tie provides no comparison. The pair-level decision rate in Supplementary Fig. S19a is one minus the fraction of tied pairs. It is distinct from the group-level tie rate in panel b. Accuracy is calculated only for usable pairs; ties count as neither successes nor errors. The Top-1 hit rate is the expected fraction of experimentally recorded entries among structures tied for the criterion's minimum. Its random baseline is the experimentally recorded fraction in that group. The archived ranking table reports Top-1 lift alongside decision rate. Set 4 distinguished 28.2% of all pairs and selected no unique structure in 85.1% of composition groups.

Every statistic is first calculated within a composition. We then average with equal weight across compositions for which that criterion yields a usable comparison. The number of contributing groups therefore differs among criteria, and their accuracies do not share one set of pairs. Explicit tie handling and group-equal weighting are both essential.

In R10, counting 77.7% of ties as errors produced the false claim that Pauling's rules perform worse than chance. In R9, pair weighting allowed SiO_2, which supplies 43.6% of all pairs, to dominate the comparison. Under pair weighting, density versus formation-energy accuracy was 0.7103 versus 0.6476. Group-equal weighting reversed it to 0.6347 versus 0.7451.

Confidence intervals come from a cluster bootstrap that resamples composition groups with replacement and recomputes the group-equal statistic. This unit preserves within-composition correlation. Pair-level resampling would narrow the intervals to a meaningless width. Folds are split by composition group. The bootstrap used $B = 500$ replicates and seed 20260728.

S14 Supplementary figure conventions and source data

All supplementary figures are drawn in Arial and carry no in-panel titles. The subject of each figure is stated in the first clause of its caption. Source data accompany the paper, and the plotting code performs no computation.

S15 PRIS-derived synthesis score for composition-held-out ranking

This note documents the score reported in the main text as the PRIS-derived synthesis score (PSS).

Why a new analysis. The plausibility laws were fitted to damage detection, whereas synthesizability screening benefits from a continuous ordering of structures within the same composition. PSS was therefore fitted directly to the experimental-record label under the composition-held-out protocol documented here.

Connection to the wider programme. This analysis and the refit in Section S16 belong to the autonomous programme documented in Section S1. Both were executed in sessions dated 2026-08-14, after the first authorised reserve evaluation. They reuse the programme's data, feature conventions, and evaluation code. The unified statistics in Section S1 include both analyses.

Pre-registration. The synthesis-score plan was fixed before fitting, and its SHA-256 was recorded in the repository. Revision 1 added a minimum decision rate of ≥ 0.99 after development search found a high-accuracy solution that ranked too few pairs. Revision 2 added a secondary hybrid model with its form fixed before evaluation. Both revisions predated the single held-out evaluation. Before computing any new row, we reproduced four published baselines to four decimal places: Pauling 5, ρ, volume per atom, and DFT E_{hull}.

Split. A seeded hash of each composition string assigned all 1,508 two-class groups to development or held-out sets. Development contained 919 groups, 4,187 structures, and 14,563 pairs. Held-out evaluation contained 589 groups, 2,571 structures, and 4,357 pairs. SiO_2, which supplies 0.436 of all pairs, fell in the development set. The held-out results are therefore free of the known single-composition dominance in R9. The evaluation script accessed the held-out set exactly once, after every coefficient was fixed and hashed.

Feature pool. We began with 85 stored structural features in the task table. The pre-registered exclusion list removed extensive counts, Ewald totals, and a table-availability artefact. Nine features were then added by physical function. Symmetry features comprised space-group number, crystal-system rank, and distinct-site fraction at two tolerances. Classical-repulsion features were $\sum(r_{\mathrm{sum}}/d)^9/N$ and $\sum\exp[(r_{\mathrm{sum}}-d)/0.345\,\text{Å}]/N$, resolved by charge-sign pair. The remaining additions were cation–anion elastic strain and mass density. No feature uses DFT output. Charges are composition-only integer valences, and radii are Shannon radii, as throughout this work.

Model class. We used an antisymmetric pairwise logistic score with no intercept. Reversing a pair therefore reverses the score without changing its magnitude. Features were standardised using development data, and missing values were handled by the frozen preprocessing statistics. Pairs received equal weight within each composition. Greedy forward selection used five-fold grouped cross-validation, at most eight terms, and a minimum gain of 0.002. The final model was refitted on all development data. Its coefficients and preprocessing statistics were stored in a SHA-256-hashed JSON file before held-out evaluation.

Development result. Forward selection added terms in this order: (1) $\overline{E_{\mathrm{M}}/|z|}$, score 0.6299; (2) distinct-site fraction, 0.6870; (3) mean relative bond-valence deviation, 0.7214; (4) volume per atom, 0.7344; (5) maximum polyhedral connectivity, 0.7442; and (6) fraction of isolated polyhedra, 0.7548. A seventh candidate gained +0.0017, below the pre-registered minimum, and was not retained.

An L2-regularised model using all allowed features scored 0.7572. DFT hull energy scored 0.7500 on the same development groups. The archive contains the fixed coefficients, preprocessing statistics, SHA-256 hashes, and complete single-feature audit. Leading single features were distinct-site fraction (0.6837 at decision rate 0.79), the bond-valence family (0.664–0.667 at decision rate ≥ 0.99), and density (0.6495).

Experimental records. The synthesis task uses the same classification of experimentally recorded structures and structures without an experimental record as Section S13.

On held-out data. Across 589 held-out groups, PSS scored 0.6811 at decision rate 1.000. The full-feature-pool variant scored 0.7059. The hybrid with hull energy, whose form was fixed before evaluation, scored 0.6865. Reference scores were 0.7503 for DFT hull energy, 0.6346 for volume per atom, 0.6219 for Shannon packing, and 0.5597 for ρ. Paired cluster-bootstrap differences were -0.0646 $[-0.1043, -0.0267]$ for PSS minus hull and -0.0592 $[-0.0963, -0.0234]$ for hybrid minus hull.

PSS decided every group and outperformed every single structural score. Its Top-1 lift was $+0.2003$, compared with $+0.2971$ for hull energy. Neither PSS nor the hybrid overtook hull energy overall. As specified before evaluation, the main text reports that PSS does not match DFT accuracy and gives the difference with its interval. The decline from 0.7548 in grouped cross-validation to 0.6811 on held-out compositions measures optimism from repeated development. The log records exactly one held-out evaluation.

Accuracy among confident comparisons. This analysis was performed after held-out evaluation. For each criterion, pairs were ordered by $|\Delta s|$, the absolute pairwise score difference. The table reports pair-level accuracy and cluster confidence intervals; the complete resolution table is archived.

pairs kept	PSS	95% CI	E_{hull}	paired Δ [CI]
1.00	0.7384	[0.708, 0.777]	0.8074	—
0.50	0.8237	[0.785, 0.851]	0.8476	-0.024 $[-0.082, +0.062]$
0.30	0.9135	[0.863, 0.944]	0.8187	$+0.095$ $[-0.005, +0.139]$
0.20	0.9437	[0.896, 0.969]	0.8439	$+0.100$ $[+0.012, +0.199]$
0.10	0.9655	[0.905, 0.988]	0.9241	$+0.041$ $[+0.000, +0.217]$

The retained-pair fractions were examined after the single held-out evaluation; this breakdown is descriptive rather than pre-registered. At the most confident fifth, $CeSe_2$ supplies 0.245 of pairs. The all-pair values differ from the reported group-equal comparison of 0.6811 versus 0.7503 because held-out groups still differ in size, even without silica.

Only the most confident fifth has a paired interval excluding zero. A Bonferroni-adjusted 99 per cent interval across the five examined fractions still excludes zero ($+0.100$ $[+0.004, +0.220]$). Removing the most represented composition strengthens the difference to $+0.159$ $[+0.033, +0.213]$. At retained fractions of 100%, 50%, 30%, 20%, and 10%, the largest-composition shares are 0.331, 0.120, 0.174, 0.245, and 0.450, respectively. This remains an exploratory breakdown of one held-out evaluation.

Stability of the coefficients and correlation of the terms. The six terms remained fixed; only their coefficients were re-estimated. The feature search was not reopened. Every result below uses only the 919 development groups, comprising 4,187 structures and 14,563 within-composition pairs, under frozen preprocessing statistics. The held-out groups were not evaluated again.

Refitting random subsets of development structures moved individual coefficients but barely changed the score ranking. Across 200 draws at each size, mean Spearman agreement with the published score was 0.973 on half the structures and 0.998 on nine tenths. Mean cosine similarity of the coefficient vectors was 0.983 and 0.998, respectively.

Coefficient magnitudes were less stable and varied by term. On half the structures, the 95% range spanned ±21% of the published $\widetilde{\eta}_{\mathrm{site}}$ coefficient but ±116% of $\widetilde{M}_z$. Every coefficient retained its published sign in at least 95.5% of draws (Supplementary Fig. S15a,c).

A cluster bootstrap over composition groups used all development data and the interval convention of Section S13. It produced the same term ordering: $|\beta|/\mathrm{SE}$ ranged from 12.1 for $\widetilde{\eta}_{\mathrm{site}}$ to 2.0 for $\widetilde{M}_z$. Every coefficient retained its sign in at least 99.5% of resamples (Supplementary Fig. S15b). The development data fix the signs and relative weights; an individual coefficient should not be read to its second decimal.

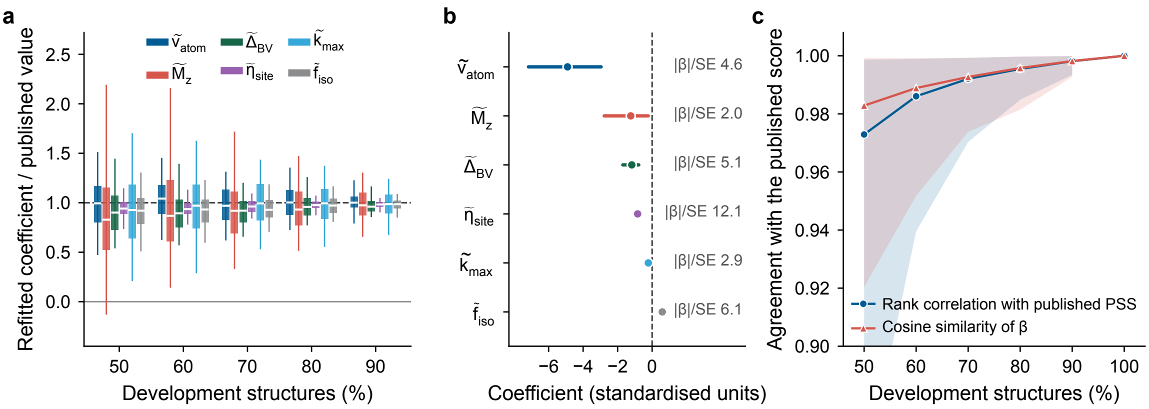

Fig. S15 Coefficient stability of the fitted synthesis score under resampling of the development set. The six terms are held fixed and only their coefficients are re-estimated. **a**, Refitted coefficients divided by their published values for 200 random subsets at each size. Boxes span the interquartile range, white bars mark medians, and whiskers span the 2.5th to 97.5th percentiles. The dashed line marks the published value. **b**, Cluster-bootstrap coefficient intervals over all development composition groups ($B = 2{,}000$). Intervals span the 2.5th to 97.5th percentiles. Circles mark published values, and right-hand labels give each coefficient divided by its bootstrap standard error. **c**, Agreement between each refitted score and the published score. The two metrics are Spearman rank correlation across development structures and cosine similarity between coefficient vectors. Lines give means; shading spans the 2.5th to 97.5th percentiles.

The six terms are correlated without being redundant. Across structures, the strongest Spearman correlations are -0.51 between $\widetilde{k}_{\max}$ and $\widetilde{f}_{\text{iso}}$, and 0.50 between $\widetilde{v}_{\text{atom}}$ and $\widetilde{M}_z$. The pairwise fit instead uses within-composition differences. In that space, every correlation is below $|0.36|$, every variance inflation factor is at most 1.29, and the design correlation matrix has condition number 3.58 (Supplementary Fig. S16). The coefficient spread in Supplementary Fig. S15a therefore reflects the number of independent composition groups, not near-duplication among terms.

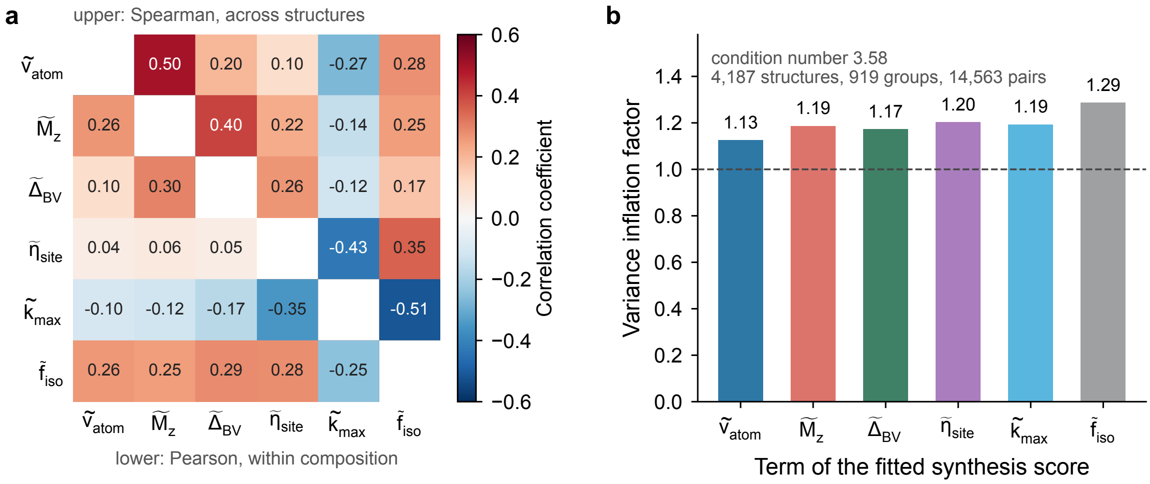

Fig. S16 Correlation among the six terms of the fitted synthesis score. **a**, Correlation matrix of the standardised terms on the development set. Entries above the diagonal are Spearman correlations across structures; entries below it are Pearson correlations of the within-composition differences that the pairwise fit uses. **b**, Variance inflation factors of the same within-composition design, with the dashed line marking the uncorrelated value of one. The condition number of the design correlation matrix is given in the panel.

S16 PRIS and PSS track predicted synthesizability across two PU models

Positive–unlabelled (PU) learning lineage and expanded pool. Jang et al. introduced bagged PU-CGCNN and the crystal-likeness score (CLscore). Their training data comprised 46,781 ICSD-linked Materials Project positives and 77,734 virtual unlabelled structures [1]. A later study applied the pretrained CLscore model to 1,401,562 theoretical structures. It selected 80,000 low-CLscore candidates for a crystal-synthesizability language model [2].

Here, we expanded the positive set to 99,162 structures: 73,823 from ICSD and 25,339 from COD. We expanded the unlabelled pool to 8,125,976 LeMat and ELEMENTA candidates [3, 4].

We retrained a 50-bag CGCNN-PU model and fitted 50 MLP PU heads to frozen 128-dimensional MatterSim-v1.0.0-1M representations [5]. The two models agreed on 364,771 low-CLscore records.

Exact-CIF deduplication removed 179 repeats, leaving 364,592 unique PU-selected hard-negative structures for analysis.

Full-pool comparison. Set 1–Set 4 were applied as independent law-set choices. Set 1 gave 94.8% experimental-structure satisfaction and 0.2% hard-negative screening. The corresponding values were 88.3% and 1.8% for Set 2, 87.1% and 1.9% for Set 3, and 80.7% and 51.9% for Set 4. PSS was evaluated as a continuous threshold curve. At 80.7% experimental-structure satisfaction, it screened 83.7% of the hard-negative pool.

All six PSS descriptors were observed for 21,477 experimental structures and 135 hard negatives. For all other records, unavailable descriptors were replaced with medians fixed on the PSS training data before this analysis.

Mechanism attribution. Fractional attribution preserves the logical OR that defines Set 4. A structure violating k laws contributes $1/k$ to each violated law, so the shares sum to the complete violation rate. A complementary leave-one-law-out calculation removes one law and recomputes both experimental-structure satisfaction and hard-negative screening.

Law 7 supplies 50.28 percentage points of the 51.9% hard-negative screening rate. Removing it loses 49.49 percentage points of screening while recovering 5.00 percentage points of experimental-structure satisfaction. Law 1 contributes more strongly to the controlled-damage benchmark. Different mechanisms therefore dominate the two tasks (Supplementary Fig. S17a,b).

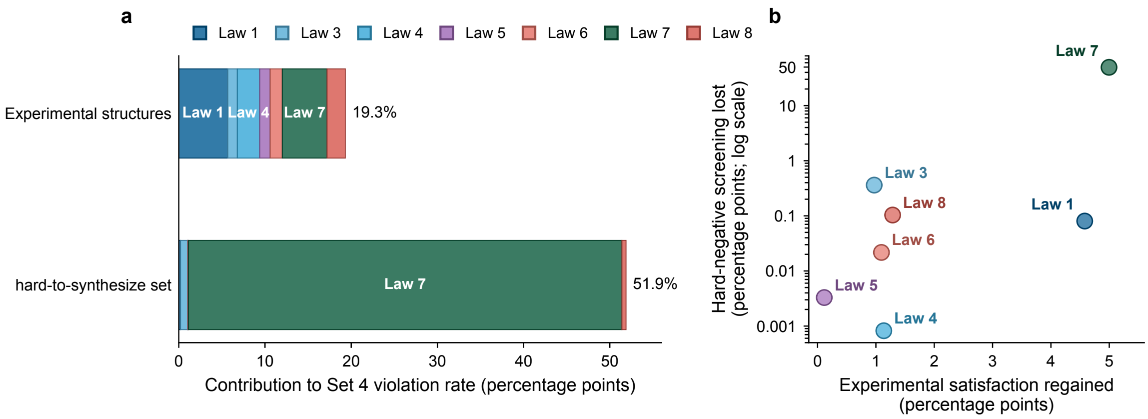

Fig. S17 Law-wise attribution of Set 4 screening in experimental and PU hard-negative structures. Records without a verdict remain in both cohorts. **a**, Stacked bars allocate the Set 4 violation rate among Law 1 and Law 3–Law 8 for the deduplicated experimental and PU hard-negative cohorts. Each multi-law violation is divided equally among its active laws. **b**, Each coloured point is a leave-one-law-out comparison, with experimental-structure satisfaction regained on the horizontal axis and hard-negative screening lost on the logarithmic vertical axis.

Two PU representations. CGCNN-PU is a task-trained crystal graph convolutional network. MatterSim-1M-MLP-PU uses frozen 128-dimensional representations from the MatterSim-v1.0.0-1M checkpoint followed by an MLP head, providing a complementary encoding learned during universal-potential training. Both models were evaluated across 50 validation bags containing held-out experimental positives and sampled unlabelled pseudo-negatives. Mean ROC–AUC was 0.976538 ± 0.000763 for CGCNN-PU and 0.946566 ± 0.000942 for MatterSim-1M-MLP-PU. Each CGCNN-PU bag contained 19,798 positives and 19,798 pseudo-negatives. MatterSim embedding availability left approximately 37,850 records per bag. Supplementary Fig. S18a–d shows the mean ROC curves, bag-wise envelopes, and confusion matrices at the fixed 0.5 decision threshold.

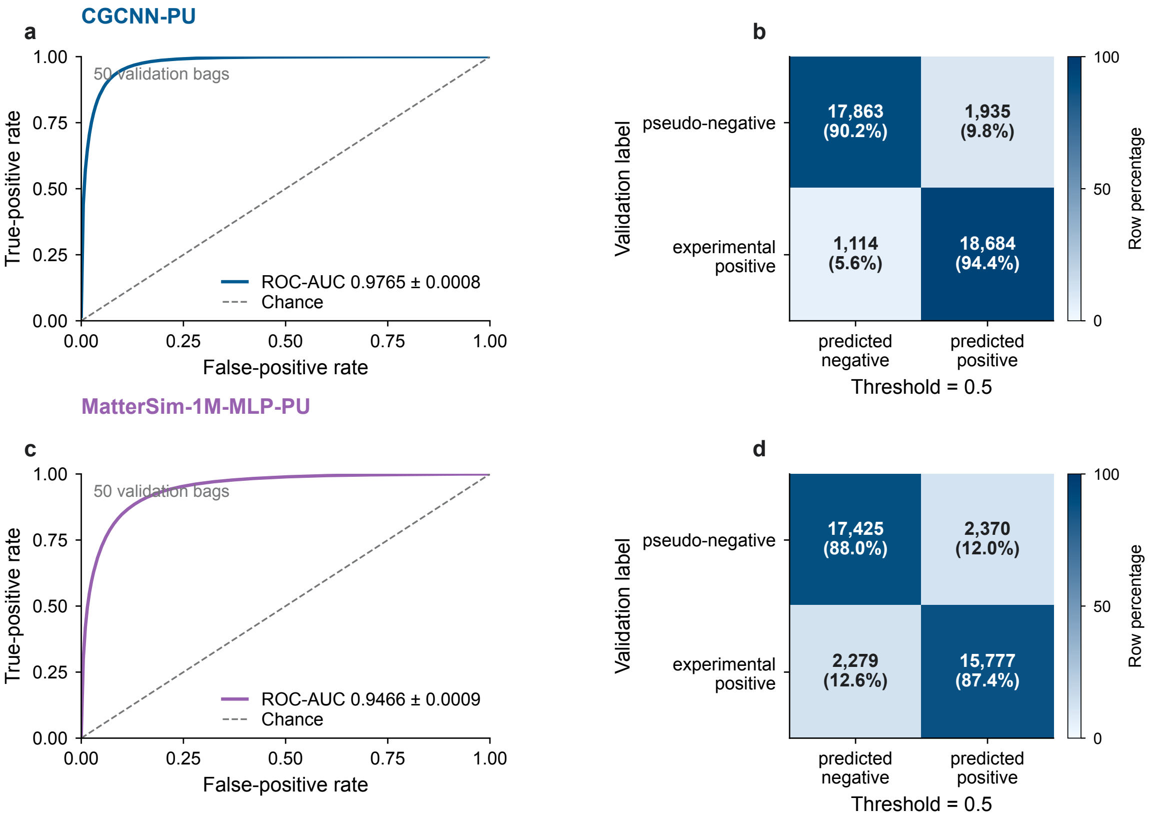

Fig. S18 Validation performance of the two positive–unlabelled learning models. Every bag contains held-out experimental positives and sampled pseudo-negatives. **a,c**, Mean receiver-operating-characteristic curves for CGCNN-PU and MatterSim-1M-MLP-PU across validation bags. Axes are false- and true-positive rates, shading gives the central bag-wise envelope, legends report the bag-level ROC–AUC summary, and the diagonal marks chance. **b,d**, Confusion matrices at the fixed decision threshold. Rows are validation labels, columns are predictions, and cells show bag-mean counts and row percentages.

Continuous CLscore relation. Raw scores from the two models have different scales and were not averaged directly. Records were first assigned to ten within-model percentiles. PRIS and PSS summaries were calculated separately, then averaged pointwise across aligned percentiles. From the

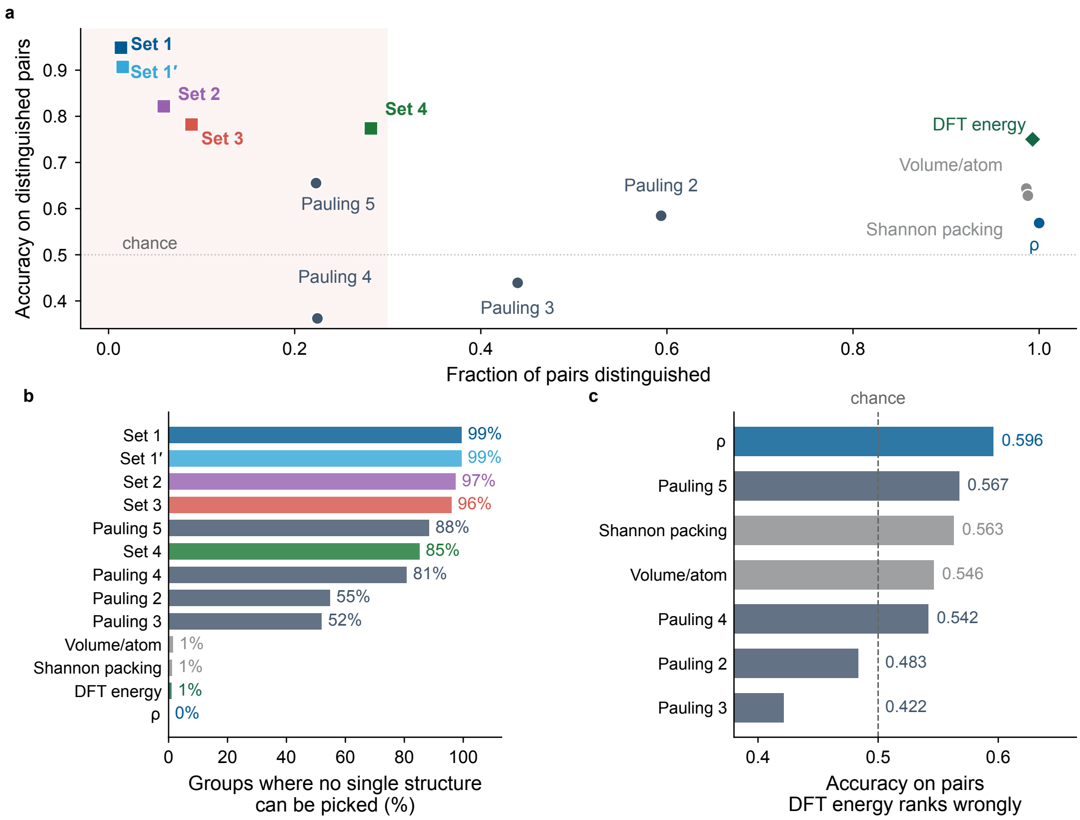

Fig. S19 Decision boundaries in same-composition polymorph ranking. a, The fraction of within-composition pairs distinguished is plotted against group-equal accuracy on distinguished pairs. Shapes and colours identify baseline criteria, PRIS law sets and DFT energy, and the shaded band marks the low-decision region. **b**, Bars show the fraction of composition groups for which no unique structure is selected. **c**, Bars show non-energy-criterion accuracy on pairs ordered incorrectly by DFT E_{hull}. Ties are excluded from accuracy, and horizontal lines mark chance.

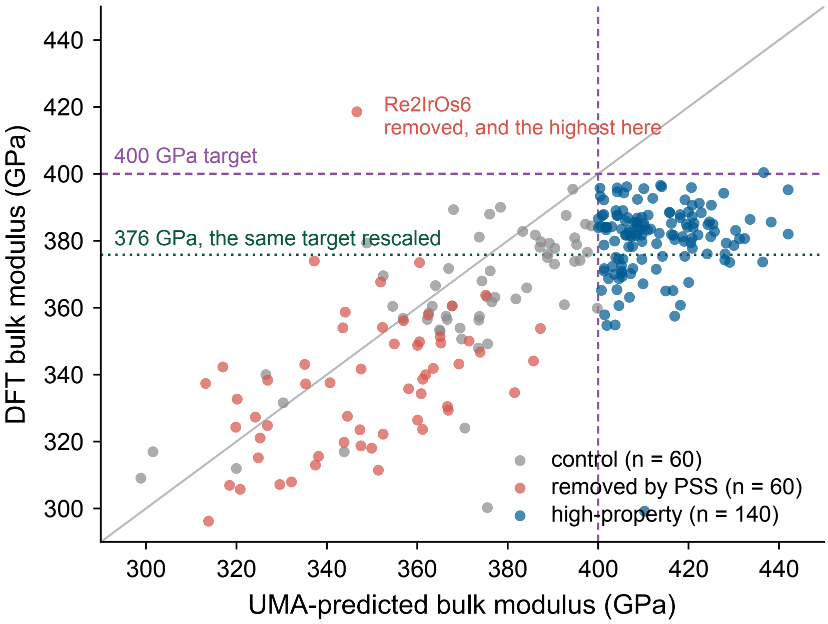

Fig. S20 The property screen against first-principles bulk moduli. Each point is one candidate. The horizontal axis is bulk modulus predicted by the UMA machine-learning model, which defined the design target. The vertical axis is the DFT bulk modulus from a third-order Birch–Murnaghan fit to a five-point energy–volume curve. The plot shows 260 of 261 selected candidates; one cell did not converge. The DFT values are a median 0.940 times the machine-learning predictions, so the model predicts systematically higher bulk moduli. Thus its 400 GPa target (dashed) maps to 376 GPa on the DFT scale (dotted). Ranking is nevertheless retained: a high-property candidate outranks a removed candidate 0.966 of the time. Only three of the 140 highest DFT bulk moduli were removed by the screen. Re_2IrOs_6 is the informative exception: the screen removed it, but DFT ranks it highest. This cohort deliberately oversamples high machine-learning predictions, so these fractions do not transfer to the full candidate pool.

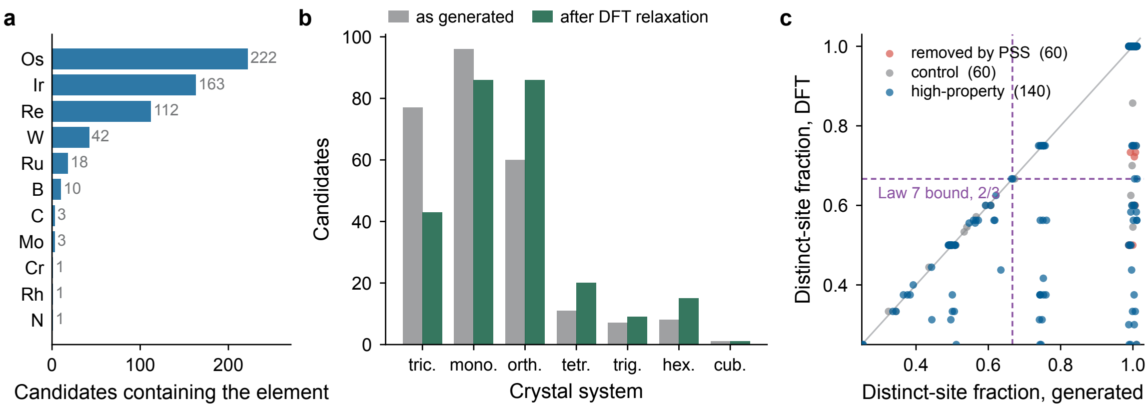

Fig. S21 What the property-conditioned queue is made of, before and after relaxation. The panels show all 260 candidates whose bulk moduli were computed. **a**, Number of candidates containing each element. Conditioning on a bulk modulus above 400 GPa drives the generator into a narrow corner of the periodic table, dominated by the 5d transition metals. **b**, Crystal system before and after DFT relaxation. Both use the 0.01-Å tolerance applied in every Law 7 verdict. **c**, The Law 7 distinct-site fraction before and after relaxation. Colour gives the synthesizability-screen decision. Points can only fall below the diagonal because relaxation merges sites and does not split them. Dashed lines mark the Law 7 threshold of two-thirds on each axis. Relaxed cells are provided as Supplementary Data.

lowest to highest percentile, mean Set 4 violation fell from 50.3% to 30.4%, while mean PSS rose from -11.37 to -0.18. The shared direction across task-trained and pretrained representations is the cross-model evidence used in the main text.

MatterSim-computed E_{hull} threshold. A separate balanced pilot sampled 1,024 experimental structures and 1,024 PU hard negatives. MatterSim relaxation returned finite energies for 2,045 structures. For each composition, we subtracted the Materials Project phase-hull reference. We then swept the E_{hull} threshold. At 0.20 eV per atom, the method retained 86.2% of the experimental sample and screened 72.0% of the PU sample. The energy route therefore requires relaxation and phase-reference construction beyond direct PRIS evaluation.

Property-conditioned inverse-design protocol. We ran MatterGen v1.0.3 from model-repository commit `67afb22ea0dd0073aedfc85b30b94bea0ee21863`. The public bulk-modulus checkpoint conditioned 13 independently seeded shards at 400 GPa, yielding 1,664 parsed structures.

StructureMatcher deduplication used `ltol=0.2`, `stol=0.3`, and `angle_tol=5` degrees. It removed 583 equivalent outputs and left 1,081 unique, unrelaxed structures.

The `uma-s-1p1` checkpoint independently predicted bulk modulus through its `omat` energy task. We evaluated energies at fixed fractional coordinates and five cell volumes from 0.96 to 1.04 times the generated volume. Each quadratic $E(V)$ fit had to reach $R^2 \geq 0.98$. All 1,081 fits passed, and 140 structures had a clamped machine-learning prediction of at least 400 GPa.

PSS coefficients and preprocessing remained fixed from Supplementary Section S15. Each generated structure supplied two of six PSS descriptors; the other four used frozen training medians. We calibrated the threshold on 541 experimentally recorded structures with the same two-descriptor support and a valid UMA prediction of at least 200 GPa. A target satisfaction of 97.5% selected PSS cutoff -0.636879, which retained 528 of the 541 calibration structures.

Scores below this cutoff were screened, whereas ties were retained. The cutoff produced 97.6% satisfaction, removed 61 generated candidates, and retained all 140 high-property candidates. No generated score or property label entered fitting, preprocessing, or cutoff estimation. The generated queue defined only the matched two-descriptor support stratum. Within this stratum, the frozen score reduces exactly to

$$\mathrm{PSS} = 5.597386 - 3.133285\,\eta_{\mathrm{site}} - 0.206563\,v_{\mathrm{atom}}.$$

All 61 screened structures had $\eta_{\mathrm{site}} = 1$, passed both the 0.5- and 0.7-Å distance cutoffs, and violated Law 7. Their volumes ranged from 15.017 to 16.558 Å^3 per atom, with mean 15.329 Å^3. The retained queue had mean 14.331 Å^3 per atom. The PSS-screened structures were a strict subset of the 728 Set 4 violations. Set 4 also removed 95 of the 140 high-property candidates, whereas denser packing kept them above the continuous PSS threshold.

The matched Ir_2Os_7 examples in the main figure make this distinction concrete. The screened P1 structure has 18 inequivalent sites among 18, $v_{\mathrm{atom}} = 15.017$ Å^3, and PSS $= -0.638$. The retained C2/m structure has four inequivalent sites among nine, $v_{\mathrm{atom}} = 14.327$ Å^3, PSS $= 1.245$, and a UMA-predicted bulk modulus of 408.7 GPa.

Generation manifests, model hashes, and the deduplicated archive are preserved as generation/model artefacts. Evaluation tables and the final support-matched evaluation are preserved as analysis artefacts.

S17 Extending the plausibility sequence with structural simplicity and bond valence

Pre-registration. The Set 4 plausibility plan was fixed before the augmented matrix features or any law search existed. Its SHA-256 was recorded in the repository. The analysis reused audited, reserve-free tables containing 17,929 experimental and 12,202 damaged structures. Damaged structures were recreated with the pipeline's original seeds. Before search began, Set 3 was reproduced to all four published decimals on both splits.

The model class kept Set 3 unchanged and allowed at most four added laws. Each addition had to be expressible as a one-line physical statement. Candidate thresholds came from a fixed percentile grid of the experimental discovery distribution. This produced 8,466 candidate laws over 94 structural quantities that met the inclusion criteria. Greedy selection maximised discovery damage detection while requiring experimental satisfaction of at least 0.81.

The prespecified script evaluated the held-out split once. Its earlier repeated use is disclosed in Section S8.

Selection. The search stopped at its prespecified minimum required gain after two additions:

- **Law 7 (simpler-structure law):** the fraction of crystallographically inequivalent sites is at most 2/3. This writes Pauling's preference for simpler structures as a threshold law. The threshold is the 90th-percentile grid point of the experimental discovery distribution, or about the 92nd percentile of structures.

- **Law 8 (bond-valence deviation):** the mean relative deviation of bond-valence sums from nominal valences is at most 0.7143. This hard threshold expresses the electrostatic valence principle.

On discovery data, satisfaction was 0.8107 and damage detection was 0.9051. A third candidate improved detection by only 0.0035. This gain was below the pre-registered minimum, so selection stopped.

On held-out data. Satisfaction was 0.8180 and overall damage detection was 0.9111. Detection by type was 0.7338 for D1, 0.9091 for D2, 1.0000 for D3, 0.9288 for D4, and 0.9851 for D5. Of 5,297 experimental structures, 3,713 satisfied Set 4, 964 failed at least one law, and 620 received no verdict. Of 3,612 damaged structures, 237 satisfied Set 4, 3,291 failed at least one law, and 84 received no verdict. All three pre-registered criteria were met: satisfaction ≥ 0.80, damage detection ≥ 0.80, and no type below 0.55. The lowest type-specific detection exceeded its required minimum by 0.18.

Testing on one damage type omitted during fitting. The five unseen-type detection rates were 0.7243 for D1, 0.8924 for D2, 1.0000 for D3, 0.7928 for D4, and 0.6863 for D5. The folds omitting compression, cation–cation swap, or displacement selected the same two laws at the same thresholds. The folds omitting expansion or cation–anion swap selected different additions. They still reached 0.79 and 0.69 detection on their unseen types, respectively. This behaviour is consistent with physicochemical laws rather than one perturbation-specific pattern.

The 0.2318 result for the provably optimal depth-3 tree is not a direct comparison. This test refits additions on top of fixed Set 3, whereas the tree was refitted in full.

A further evaluation population. Without refitting, we applied Set 4 to the main-text Fig. 4a,b benchmark. It contains 440 experimental parent structures and 2,024 damaged counterparts generated with the Section S12 operators. The parents came from the analysis set. Counts across the three partitions were 259/115/66, so this population is not disjoint from the search splits.

Set 4 achieved satisfaction 0.8295 and damage detection 0.8790. Type-specific detection was 0.7295 for D1, 0.8636 for D2, 1.0000 for D3, 0.8045 for D4, and 0.9909 for D5. These values are consistent with the held-out result. Overall detection was 27 times that of the 0.7-Å filter on the same structures. Per-structure results accompany the figure source data.

S17.1 Preference for simpler structures and bond valence detect different failure types

Applied singly on top of fixed Set 3, the additions divide the work. This is a descriptive decomposition of the published Set 4 evaluation, not a new success test. Set 3+Law 7 achieved held-out satisfaction 0.8361 and damage detection 0.8696. Its D1–D5 rates were 0.678, 0.899, 1.000, 0.788, and 0.984. Set 3+Law 8 achieved satisfaction 0.8964 and detection 0.7622. Its corresponding rates were 0.687, 0.596, 0.960, 0.915, and 0.609.

Law 7 provides most of the gain for swaps and displacements. Law 8 improves expansion detection and gives smaller improvements elsewhere. Law 7 detects nearly every perturbation that breaks crystallographic equivalence, but few other damaged structures. R5 identified the same class sensitivity when symmetry was tested alone. Within Set 4, other laws detect the classes Law 7 misses. The combined model provides a numerical form of Pauling's fifth rule. The experimental merge control below confirms that merging does not generally erase distinct sites in experimental structures.

Law 7 also depends on a symmetry tolerance. Site orbits are identified on the as-given cell with spglib at `symprec=0.01`. Increasing this tolerance to 0.1 changes verdicts for only 0.15 per cent of experimental and 0.26 per cent of damaged structures. The law presumes symmetrised input; structures with refinement noise should first be symmetrised.

When D5 was omitted, the refit selected a related symmetry law, space-group number ≥ 2, rather than Law 7 itself. Thus the data repeatedly select the symmetry and distinct-site family, not one exact threshold.

S17.2 The bond-valence law isolates extreme values

Across 10,725 experimental discovery rows with finite values, mean relative bond-valence deviation had median 0.094, upper quartile 0.200, ninetieth percentile 0.524, and ninety-fifth percentile 0.682. The fixed threshold 0.7143 is the 96th-percentile grid point. The same rows had median global instability index 0.215 valence units, close to the ~ 0.2 v.u. convention in bond-valence practice.

Law 8 is therefore a deliberately permissive upper-tail bound, not a working-chemist's quality criterion. Its discrimination comes from distribution separation. Across finite discovery and held-out rows, the medians were 0.094 for experimental and 0.705 for damaged structures. The law excludes the extreme tail where damaged structures concentrate. Rows with no computable bond-valence parameters (2,698 of 17,929) satisfy Law 8 automatically under the missing-value convention of Section S3.3.

S17.3 The physical families recur across search settings

The prespecified greedy search was repeated at required satisfaction values 0.75, 0.78, 0.81, 0.84, and 0.87. Discovery damage detection remained between 0.85 and 0.95 throughout. High detection is therefore not an artefact of the 0.81 requirement, which primarily sets satisfaction.

The exact Law 7+Law 8 pair appeared only at 0.81. At 0.78 and 0.84, the search again selected the Law 8 quantity, but at a tighter threshold. At three of the four other satisfaction levels, a space-group law replaced Law 7. This recurrence identifies a symmetry and distinct-site family rather than one fixed threshold.

When all Pauling-derived families were excluded, contact and charge-topology laws still achieved 0.8655 detection at the same satisfaction. This retains about 80 per cent of the gain over Set 3. The excluded features comprised Wyckoff, space-group, bond-valence, and rule-2/4/5 quantities. When only symmetry features were excluded, detection reached 0.8810 and the bond-valence family was selected first.

Pauling-derived families are therefore not necessary for high damage detection on this suite. A free search nevertheless selects them first, and they provide the final improvement.

S17.4 The full law-set sequence preserves its advantage at matched satisfaction

The 27-fold margin over the 0.7-Å filter compares settings with different satisfaction. For a matched comparison, an absolute cutoff of 1.71 Å was tuned to Set 4's 83.0 per cent satisfaction. It detected 0.339 of damaged structures, compared with 0.879 for Set 4, a 2.6-fold margin. The corresponding matched-satisfaction margin for Set 1 was 2.1-fold.

The parent-cluster bootstrap interval for Set 4 satisfaction was $[0.793, 0.864]$, centred at 0.830. Its damage-detection interval was $[0.864, 0.894]$, centred at 0.879. Among 66 reserve-partition parents, satisfaction was 0.879 and damage detection was 0.909. The 0.7-Å baseline detected 0.026. These reserve-parent values are not an untouched reserve test, because the S8 split-labelling error exposed part of the wider reserve to a full-sample threshold fit.

S17.5 The law-set sequence responds smoothly to increasing perturbation amplitude

The Section S12 benchmark uses one amplitude per damage type. We therefore repeated three perturbations at graded amplitudes on the same 440 parents (Supplementary Fig. S5). The PRIS sequence begins responding near 10 per cent strain and 0.1 Å displacement. By contrast, the fixed 0.5- and 0.7-Å distance filters remain near zero detection below 0.4 Å displacement.

Under Gaussian displacement, Law 7 reaches 1.00 detection by 0.05 Å because any Cartesian noise breaks Wyckoff degeneracy. The simpler-structure law thus responds first at small amplitudes. Set 1–Set 3 show a progressive response over the same range.

S17.6 Law sets fixed before evaluation separate generator outputs beyond distance filters

The analyses above contain no generative-model samples. We therefore asked whether the fixed law sequence separates raw generator outputs beyond distance filters. The sample comprised 500 MatterGen structures from the deposited, seeded seven-generator benchmark below. Every structure satisfied both fixed absolute-distance filters at 0.5 and 0.7 Å.

Across all 500 outputs, the fractions receiving a verdict and satisfying each set were 32.4 per cent for Set 1, 30.8 per cent for Set 2, 26.8 per cent for Set 3, and 0.8 per cent for Set 4. No-verdict cases remain in these denominators but are not classified as failures. For Set 3, 134 structures satisfied the set, 11 failed it and 355 received no verdict.

Even with the mean-valence fallback, 338 structures had no charge assignment. Among 162 charge-assignable structures, 4 satisfied Set 4, 155 failed it, and 3 had no verdict because another measurement was unavailable. None of the 162 failed the reduced-distance law. The later separation therefore comes from chemical, electrostatic, and site-order laws.

These raw outputs have no DFT results, so law satisfaction does not establish correctness. The interatomic-potential analysis below provides an energy check. Charge assignment remains the primary limit on verdict coverage for new structures.

S17.7 The distinct-site law exposes excess site complexity in a public generative release

Protocol fixed before computation. We downloaded all 554,054 structure files in the public GNoME stable-structure release from its official repository. Before parsing any structure, a written protocol fixed the seed for a uniform sample of $N = 5{,}000$.

For each primitive cell, spglib identified site orbits at `symprec=0.01`. The deposited space-group label was never used in the verdict. This primitive-cell convention differs from the as-given-cell convention in the discovery tables. Absolute external Law 7 rates, and Set 4 rates containing Law 7, are therefore descriptive rather than directly comparable with the discovery rate. Every sampled structure was parsed and analysed for symmetry.

Result. Law 7 was not satisfied by 2,032/5,000 = 40.6% of the sample. By deposited label, failure rates were 100% for P1 (727/727), 96.6% for Pm (731/757), and 88.4% for Cm (305/345). Rates were 16.6% for Amm2, 1.4% for C2/m, and 7.7% for all other groups pooled.

Orbit detection might assign higher symmetry than the deposited label and thereby change verdicts, especially for P1. We therefore repeated the analysis at the tenfold coarser `symprec=0.1`; the failure rate changed by less than one percentage point, to 39.7%. At the primary tolerance, spglib assigned a strictly higher group than the deposited label for only 0.02% of the sample and agreed with the label for 97.5%.

Failures are therefore concentrated in P1, Pm, and Cm, quantifying the release's published space-group anomaly. The sample's median distinct-site fraction is 0.60, compared with the experimental 90th-percentile threshold of 2/3. The protocol, per-structure table, and summary are archived.

The similar-element merge test. The over-ordering hypothesis is that the release assigns chemically similar atoms to distinct crystallographic sites. If so, merging same-group elements should restore symmetry in Law 7 failures. By contrast, merging should have less effect when chemically distinct sites are also geometrically distinct.

We drew a stratified sample of 300 entries from the fixed 5,000 structures. It contained 150 Law 7 failures, stratified over P1, Pm, Cm, and remaining groups, and 150 structures satisfying Law 7. Elements in the same periodic group were merged, with Sc, Y, and the lanthanides treated as one class. We then recomputed orbits and space groups under the same convention. At least one same-group pair was mergeable in 113/150 Law 7 failures and 78/150 structures satisfying Law 7. At the strict 0.01-Å tolerance, merging changed little because relaxed geometries lie off ideal sites.

We therefore used the 0.1-Å tolerance, at which the whole-sample Law 7 failure rate changes by less than one percentage point.

Among mergeable Law 7 failures, 78% moved to a distinct-site fraction of 2/3 or below and 79% moved to a higher space group. The median space group rose from Pm (no. 6) to Pnma (no. 62). Non-mergeable failures were unchanged, with median distinct-site fraction 1.0 before and after. Structures already satisfying Law 7 shifted only modestly. In this stratified sample, the intervention supports near-symmetric lattices carrying ordered labels for chemically similar elements.

An experimental control used 12,000 random structures. Only 27 both failed Law 7 at the same tolerance and contained a mergeable same-group pair. The identical protocol restored higher symmetry for none of them (0/27), and median distinct-site fraction remained 0.80. Experimental oxysulfides, oxyfluorides, and mixed alkalis place same-group elements on geometrically distinct sites, so relabelling cannot erase them. The restoration in the generative sample is thus not explained by the merge operation alone and is consistent with its element-label assignments.

A complementary database search found a disordered ICSD entry with the identical element set for only 2 of 300 sampled entries. This is consistent with the release's novelty criterion. The intervention is therefore the direct test, rather than a database lookup.

S17.8 Bond-valence and chemical-order laws dominate failures in the assignable subset

A second protocol fixed the sample and verdict procedure before all eight laws were applied to the same 5,000 structures. A final audit found that the external Law 8 implementation did not reproduce the discovery definition. We recomputed Law 8 with the discovery analysis's CrystalNN graph and tabulated bond-valence parameters. The original protocol, code, and outputs remain preserved.

Charge assignment first sought integer charge balance. If that failed, it used non-integer mean valences. Structures for which both methods fail receive no verdict and are never counted as failures. Charges were assigned to 376 structures, or 7.5% of the sample. The remainder is largely intermetallic and outside the stated domain. Within the assignable subset, individual contact and electrostatic laws had 94.7–99.2% satisfaction, as expected for a DFT-relaxed release.

law set	evaluable	satisfy	fail	no verdict
Set 3	335	281	54 (16.1%)	41
full Set 4	314	159	155 (49.4%)	62

Counts are relative to the 376 charge-assigned structures; percentages in the fail column use the evaluable denominator. Of the 155 full-set failures, 88 (56.8%) satisfied Set 3 but failed Law 7 or Law 8: 41 failed Law 7 alone, 17 failed Law 8 alone, 14 failed both, and 16 failed Law 7 while Law 8 was unavailable.

Law 8 was evaluable for 254 structures, or 67.6% of the charge-assignable subset, because coverage is limited by bond-valence parameters. It failed for 36 of these 254 structures (14.2%). Law 7 failed for 26.1% of the charge-assignable subset, compared with 40.6% of the full sample. The charge-assignable subset is more symmetric than the intermetallic-dominated remainder, and the two rates have different denominators.

S17.9 Symmetry-aware generators better preserve the distinct-site fraction

The pre-analysis protocol extended the MatterGen sample to seven public generative models and one experimental control. Structures came from the openly licensed WyckoffTransformer benchmark deposit (CC BY 4.0). We sampled 500 structures per model at a fixed seed. The control comprised 500 structures from the MP-20 test split, the standard generative benchmark of Materials Project cells containing at most 20 atoms.

The fixed absolute-distance filters had 91–100% satisfaction for every model's raw outputs and did not separate generators from the control. The law sequence did. Formal-charge requirements prevented a verdict for 66–76% of generator outputs and 57% of the intermetallic-heavy control.

Among charge-assignable outputs from coordinate-generating models without explicit symmetry constraints (MatterGen, DiffCSP, and MiAD), 0–2.5% satisfied Set 4. For symmetry-constrained models (DiffCSP++, CrystalFormer, SymmCD, and WyckoffTransformer), 29.9–46.1% satisfied Set 4. The experimental control rate was 64.3%. Structures missing another required measurement received no verdict and were excluded from these rates.

At the strict tolerance, Law 7 failed for 95–100% of raw structures from models without imposed symmetry, where nearly every atom was inequivalent. Per-structure tables and implemented thresholds are archived separately from the deposit checksum.

Robustness checks: full denominators and relaxation. First, each Set 4 rate above uses a model-specific charge-assignable subset. Law 7 requires no charges and can be evaluated on every raw output. On full 500-structure denominators, it failed for 96–100% of cells without explicit symmetry, 24–28% of symmetry-constrained cells, and 27% of controls. This reproduces the same separation and rules out charge coverage as its cause.

Second, numerically triclinic, unrelaxed outputs could make orbit detection sensitive to coordinate noise. We therefore relaxed 2,500 structures from MatterGen, DiffCSP, MiAD, SymmCD, and the experimental control. The prespecified MatterSim protocol is given in the relaxation-energy subsection below. Law 7 was then recomputed at `symprec=0.1` to tolerate residual numerical noise.

After relaxation, Law 7 failure was 65.0% for MatterGen, 67.0% for DiffCSP, 40.2% for MiAD, 30.0% for SymmCD and 26.0% for the experimental control; relaxation left the last two unchanged. Thus relaxation reduces but does not erase the Law 7 separation in these tested sets. Numerical noise in unrelaxed coordinates cannot explain the entire signal. This local MatterSim check does not establish experimental stability or a universal property of generators without explicit symmetry constraints.

S17.10 Crystal-chemistry laws complement coordinate-based validation

checkCIF/PLATON requires diffraction data, displacement parameters, and refinement fields unavailable here. We therefore did not run the established software. A protocol fixed before computation reimplemented only the four test families calculable from atomic coordinates and a unit cell:

1. missed higher symmetry, when spglib at 0.3 Å returns a higher group than at 0.01 Å;
2. short non-bonded contacts, when a non-covalently bonded pair lies below 0.85 of the van der Waals radius sum, with 0.75 and 0.95 as sensitivity values;
3. solvent-accessible voids, measured with a 1.2-Å probe on a 0.5-Å grid and flagged above 5% of cell volume, with 2% and 10% as sensitivity values; and
4. floating atoms, defined by no neighbour within 1.3 of the covalent radius sum.

These are coordinate-based reimplementations; no checkCIF/PLATON alert code is claimed or matched.

A final audit found that the external Law 8 calculation did not reproduce the discovery definition. We recomputed the values with the discovery CrystalNN graph and tabulated bond-valence parameters. The original protocol, code, and outputs remain preserved.

For the six-law comparison, each parent and damaged counterpart was converted to a primitive cell before Law 7 evaluation. Paired comparisons therefore share a cell convention. Their absolute Law 7 rates are not directly comparable with the deposited-cell discovery tables.

We took the first 300 entries with composition-only integer charge assignments from a seed-shuffled scan of 8,000 public generative structures. This enriched sample cannot estimate population charge coverage. In the uniform 5,000 sample, 222 structures (4.44%) had integer assignments.

Each parent was perturbed once per applicable Section S12 damage type. Computed parent crystals served as controls because the experimental coordinates are not redistributable. The four-check set failed for 0.303 of parents, and the six-law PRIS set failed for 0.507. After observing these parent rates, we added a paired analysis. A damaged structure counted as detected only when its parent satisfied the same method.

The six-law set comprises Law 1 and Law 4–Law 8. In this sample, Law 7 and Law 8 failed for 0.257 and 0.178 of parents. These values are comparable with 0.261 and 0.142 in the 376 charge-assignable structures above.

paired damage detection	D1	D2	D3	D4	D5
missed higher symmetry	0.003	0.045	0.000	0.000	0.000
short non-bonded contacts	0.507	0.148	0.621	0.247	0.397
solvent-accessible voids	0.000	0.000	0.000	0.037	0.000
floating atoms	0.000	0.031	0.030	0.409	0.097
four checks together	0.512	0.215	0.622	0.522	0.431
Law 7 distinct-site fraction	0.135	0.833	1.000	0.000	0.964
Law 8 bond valence	0.353	0.336	0.831	0.950	0.124
six-law set	0.697	0.892	1.000	0.936	0.982

The pre-specified comparison included structures for which both methods returned a verdict. PRIS alone detected 0.365, whereas the four checks alone detected 0.021. The post-result paired comparison additionally required the parent to satisfy both methods. The corresponding fractions were 0.430 and 0.042.

The coordinate-based symmetry check asks whether a looser tolerance reveals higher symmetry than the tight-tolerance assignment. It detected few wrong-site exchanges because it was not designed specifically for them. Law 7 instead asks whether the number of distinct site orbits is implausibly high. The unrestricted sample, both sensitivity sweeps and the per-structure table are archived with the analysis records.

S17.11 The laws return no verdict outside their domain and detect anomalies in the evaluable falsified depositions

The comparison above evaluates damage introduced by our perturbation procedures. We therefore evaluated a historical collection of falsified structures that was not created for this benchmark, without refitting any threshold. The primary source is Harrison et al. (2010), cited in the main text.

The 2010–2012 falsified depositions. Our recovered archive contains 184 structure reports listed across seven retraction notices [6–12]. Harrison et al. describe reuse of bona fide intensity data after changing one or more atoms, sometimes with cell adjustments below 4% and deleted reflections.

For 181 reports, the archive lacks structure-specific true-parent or substituted-element provenance. We therefore do not infer which depositions share one diffraction dataset. We retrieved all 184 supplementary CIFs from the journal archive because COD removes coordinates from retracted entries. Of these, 177 parsed successfully.

Molecular coordination compounds dominate the corpus and largely lie outside the laws' domain. No formal-charge assignment was available for 172/177 parsed depositions (97.2%); these receive no verdict. The one available original control exceeded the evaluation size cap and also received no verdict. These are the stated domain boundaries. All five evaluable depositions failed the laws.

Four evaluable records share an $M(C_2H_2N_3Cl)$ ligand framework and label M as Cu, Ni, Mn, or Fe in separate papers [11]. This grouping does not establish a common genuine parent. All four violate the same two laws. Their shortest contact has $\rho = 0.53$, in the strongly repulsive range because the assigned ionic radius is incompatible with the bond lengths. Mean bond-valence deviation is 0.77–0.78, above the 0.7143 threshold. The pattern matches the main-text cation-swap mechanism in falsified depositions rather than controlled damage.

The fifth evaluable structure, $KNOF_2$, fails the stricter reduced-distance law at $\rho = 0.79$. Retrieval records and verdicts are archived separately from the retraction-notice list.

S17.12 Relaxation energies provide an independent check on damage severity

A pre-analysis protocol used the MatterSim v1.0.0-5M machine-learning potential with fixed cells and FIRE optimisation. Force convergence was $0.05\,\mathrm{eV}\,\text{Å}^{-1}$ with a 200-step cap. Every relaxation converged or reached the cap.

First, seed 20260815 selected 150 random GNoME parents and counterparts produced by the paper's perturbations. Parents moved a median 0.03 Å and released less than 10^{-3} eV per atom. Median releases were 0.22 eV per atom for compression, 1.10 for expansion, and 6.05 for displacement. Expansion is a lower bound because the cell remained fixed and one third of runs reached the cap. Swap classes have only $n = 3$ and $n = 4$, because just 4/150 intermetallic-heavy parents received a charge assignment.

Second, a dated addendum fixed the MatterGen protocol before any energy was computed. All 162 charge-assignable structures satisfied Law 1 and released median 0.007 eV per atom (interquartile range 0.002–0.029; median maximum displacement 0.08 Å). The 338 structures without charge assignments released median 0.006 eV per atom (0.000–0.022).

Because every structure satisfied Law 1, the MatterGen sample cannot measure false failures or energetic separation. It supports only similarly small relaxation energies in Law 1-satisfying and no-verdict groups. The separate parent-versus-damage experiment tests perturbation severity.

S17.13 Composition and structure screens detect different failures

The comparisons above use minimum-distance filters because generative pipelines apply them as structural tests. Their reported validity metric also contains a composition test, which we evaluated on the same 440-parent benchmark. In the CDVAE lineage and LeMat-GenBench suite, this component combines SMACT-style charge neutrality and electronegativity.

Using `smact` 4.0 with default settings, 76.8% of experimental parents satisfied the composition test. It is therefore stricter on experimental chemistry than the reduced-distance law. All five perturbations preserve composition, however, so each damaged structure receives its parent's composition verdict. Conditional on a satisfying parent, measured damage detection was 0.000 for every damage type. Composition and structure screens therefore detect different failure classes. Per-structure results and the run protocol are archived with the source data.

The trade-off, stated plainly. Set 4 satisfaction is 0.818 on held-out experimental structures, and Law 7 concentrates much of the loss on low-symmetry structures. Set 4 therefore suits pipelines that prioritise damage detection; it does not replace Set 1–Set 3. Law-set choice remains a trade-off between satisfaction and damage detection, now published up to $(0.8180, 0.9111)$.

S18 Robustness and implementation details

This note retains quantitative details that the main text summarises in one sentence each.

S18.1 The reduced-distance law follows a repulsive tail, not a physical constant

The left tail of the ρ distribution over 12,601 experimental structures is steeper than exponential. Over $\rho \in [0.41, 0.93]$, a quadratic fit to the log density reduces the residual sum of squares from 11.82 to 4.31 and has positive curvature +25.7. This shape is characteristic of a repulsive wall.

The threshold 0.735 is the first percentile of the distribution, not a measured constant. A sweep shows no sharp transition near it: satisfaction is 0.9929 at $\tau = 0.70$, 0.9900 at 0.735, 0.9876 at 0.75 and 0.9728 at 0.80. Across the ten commonest anion families, Set 1 satisfaction remains at least 0.981. Phosphides are the exception in damage detection, which is 0.0330 compared with 0.27–0.32 elsewhere, although all 551 experimental phosphides satisfy the lower threshold. Thus, for phosphides, the law preserves experimental-structure satisfaction but contributes little damage detection. Satisfaction among DFT-relaxed candidates is 0.002 below that of experimental structures, so relaxation does not inflate it.

S18.2 Provably optimal trees perform well on seen damage types but lose performance on omitted types

At cost ratio 2.5, the provably optimal depth-three tree reaches satisfaction 0.9518 and damage detection 0.6732. The corresponding values for the four-law core on the same split are 0.9511 and 0.6173. Cost ratios from 1.75 to 2.25 return the same operating point. Replacing accuracy with f1-score also returns the same tree. Thus, within this search, only the cost ratio moves the trade-off curve. Depth 4 consumed 549 s without proving optimality, so every optimality claim is restricted to depth ≤ 3.

When one damage type is omitted during fitting, the tree's mean damage detection falls from 0.6712 on the fitted types to 0.2318 on the omitted type. A single ρ threshold selected by the same procedure reaches 0.2771 on the omitted type. Its five signed changes are -0.140, $+0.029$, $+0.762$, -0.336 and -0.339, so their mean conceals cancellation. Mean absolute change is 0.506 for the tree and 0.321 for the threshold. The tree nevertheless performs better on three of the five omitted types. The aggregate and per-type comparisons therefore describe different aspects of the result and must be read together. At satisfaction 0.95, the depth-three tree does not select ρ. Instead, it uses coordination-eccentricity and site-Madelung-energy-range features, whose held-out damage detection differs by damage type.

S18.3 Energy span and dominance by one composition reshape cross-database accuracy

The median intra-composition energy span is 25 meV per atom in the Materials Project experimental set, 99 in ELEMENTA, 186 in Alexandria and 449 in LeMat. A method evaluated on LeMat therefore orders structures separated by approximately 18 times more energy than those in the Materials Project set. Within the matched $[50, 200)$ meV-per-atom window, group-equal accuracies are 0.6912, 0.6323, 0.6287 and 0.5987 for MP, ELEMENTA, Alexandria and LeMat, respectively. This restriction reverses the unmatched ranking (Supplementary Fig. S3a,b).

Silica supplies 0.5932 of pairs in the MP experimental set and 0.4358 in the synthesis-target set. By contrast, the largest-group shares are 0.0283, 0.000358 and 0.0000631 in the three computational sets. With pair weighting, density outranks formation energy, 0.7103 versus 0.6476. Equal weighting of composition groups reverses the order to 0.6347 versus 0.7451, which refutes claim R9. Cross-database accuracy must therefore be reported with its energy window. Pair-weighted results on concentrated sets must also be accompanied by the largest-group share.

We later analysed all 1,508 groups descriptively, rather than as a new held-out test. With equal weight per group, the fixed PSS scored 0.726 on the full set. It also scored 0.726 after excluding SiO_2 and 0.726 after excluding every tetrahedral-framework oxide family containing Si, Al, P, Ge or B. Thus, this diagnostic is unchanged when silica or framework oxides are removed. Bare volume per atom scored 0.643 under all three conditions. The experimentally recorded member was denser in 66 per cent of the 1,507 non-silica groups. The same direction persisted in non-oxide chemistries (0.61).

A further robustness check on experimentally recorded structures. Excluding the 15 composition groups whose every experimental member is recorded in ICSD only under high pressure leaves the development accuracy of the fixed PSS unchanged (0.754 against 0.755), so the density term is not driven by high-pressure phases in the experimentally recorded class.

S18.4 Binary law sets often return ties where continuous scores can rank

Applied to synthesis ranking as binary criteria, the four law sets give the following coverage and conditional accuracy:

criterion	pair decision rate	groups without a unique choice	contributing groups	conditional accuracy
Set 1	0.0133	99.4%	39	0.9487
Set 1′	0.0152	99.3%	43	0.9070
Set 2	0.0594	97.4%	156	0.8216
Set 3	0.0892	96.0%	225	0.7822

Returning a tie is appropriate for a plausibility filter. Thresholding removes the ordering needed for ranking, which is why the main text treats ρ as a continuous quantity for that task.

S18.5 The laws and fitted scores as equations

The eight PRIS laws are written below exactly as applied in main-text Fig. 1d. Here, ρ is the reduced contact ratio from main-text equation (1), and f_i is the Pauling ionicity. The quantity E_M is the site Madelung energy in eV from an Ewald sum over formal point charges, computed with pymatgen `EwaldSummation` at its default accuracy. This is a model quantity. Its thresholds are selected from the observed distributions rather than derived:

Law 1 $\quad \rho \ \geq \ \tau, \qquad \tau \in \{0.735,\ 0.804\}$

Law 2 $\quad f_\mathrm{i} > 0.50 \implies \rho \leq 1.05$

Law 3 $\quad \overline{\mathrm{CN}}_\mathrm{an} \leq 3.333 \implies \overline{d/(r_\mathrm{cat}{+}r_\mathrm{an})} \leq 1.081$

Law 4 $\quad \mathrm{range}_i\left(E_\mathrm{M}(i)/v_i\right) \leq 31.45, \qquad v_i = |z_i|$

Law 5 $\quad \max_i E_\mathrm{M}(i) \leq 15.17$

Law 6 $\quad f_\mathrm{i} > 0.55 \implies$ no like-charge bonds

Law 7 $\quad n_\text{inequivalent sites}/n_\text{sites} \leq 2/3$ (site orbits: spglib, symprec 0.01)

Law 8 $\quad \overline{\left|\mathrm{BV\ sum} - v_i\right|/v_i} \leq 0.7143$

For the fitted PSS, $x^\dagger$ replaces a missing descriptor with the frozen development-set median. The transformed value $\widetilde{x} = (x^\dagger - \mu_x)/\sigma_x$ is then standardised using the development-set mean and spread. A positive score difference favours the experimentally recorded member of a same-composition pair.

The complete zero-intercept score from Section S15 is

$$\mathrm{PSS} = -4.90\,\widetilde{v}_{\mathrm{atom}} - 1.24\,\widetilde{M}_z - 1.18\,\widetilde{\Delta}_{\mathrm{BV}} - 0.84\,\widetilde{\eta}_{\mathrm{site}} - 0.22\,\widetilde{k}_{\mathrm{max}} + 0.59\,\widetilde{f}_{\mathrm{iso}}.$$

Here $M_z = \langle E_{\mathrm{M}}(i)/|q_i|\rangle_i$, $\Delta_{\mathrm{BV}} = \langle|\mathrm{BVS}_i - |q_i||/|q_i|\rangle_i$, and η_{site} is the number of crystallographically distinct sites divided by the total number of sites. The two topology terms are the maximum degree and isolated-node fraction of the cation-polyhedron connectivity graph.

S18.6 Phonon data separate plausibility, stability and synthesis

The phonon cohort contains 26,600 usable Materials Project structures. Of these, 1,512 come from the project's curated DFPT phonon database. The remaining 25,088 come from its 2025 finite-displacement release, computed with the pheasy lattice-dynamics framework. A protocol fixed before analysis combined three properties for each structure: energy above the hull, an imaginary-mode flag and the experimental record. For DFPT band structures, the imaginary-mode cutoff is -0.05 THz. For the finite-displacement density of states, the threshold is 10^{-4} negative weight, with threshold sensitivity recorded for each material.

Among the 1,175 materials covered by both phonon sources, cross-method flag agreement is 88.8%. Within the DFPT source, agreement between its band-structure and density-of-states criteria is 99.1%. Combining both phonon sources, 36% of on-hull structures have imaginary modes; the corresponding fraction in the DFPT subset is 14%. In addition, 42% of experimentally recorded entries are metastable. A further 4,271 structures are dynamically stable, lie within 50 meV per atom of the hull and have no experimental record. The per-law satisfaction analysis underlying the distinct-site comparison is reported in Section S18.7.

Two limitations apply. First, the phonon cohort is not a random database sample; it is biased towards small, ordered, near-hull cells. Second, the finite-displacement imaginary-mode flag is tolerance-sensitive, and most flagged experimental structures have little imaginary weight. Under a lenient tolerance, the flagged fraction of experimental structures falls from 41% to 9%. The corresponding fraction for theoretical structures falls from 61% to 31%. At every tested tolerance, experimental structures therefore carry substantially less imaginary-mode weight than theoretical ones.

S18.7 The law sets separate chemical order from energetic stability

The main text compares law-set satisfaction across energy, phonon and experimental-record classes in this cohort (Supplementary Fig. S22). All five law sets were applied to the same stored relaxed structures, for which the Law 2–Law 6 quantities had already been computed.

Formal charges follow the repository convention, using `oxi_state_guesses`, taking the first guess, assigning one state per element, and requiring at least one cation and one anion. Bond-valence analysis is not used for charge assignment. Each rate is calculated on the charge-assignable subset for that law set; structures outside that subset receive no verdict. The subsets contain 16,525, 15,975, 16,527, 16,068 and 15,310 structures for Set 1, Set 1′, Set 2, Set 3 and Set 4, respectively. On the 14,986 structures common to all five, satisfaction decreases through Set 1, Set 2, Set 3 and Set 4. Set 1′ is reported separately.

	Set 1	Set 1′	Set 2	Set 3	Set 4
satisfaction, this population	0.993	0.988	0.933	0.909	0.743
satisfaction, held-out benchmark	0.992	0.989	0.958	0.917	0.818
P(satisfies | recorded)	0.990	0.987	0.927	0.904	0.774
P(satisfies | without record)	0.995	0.990	0.937	0.914	0.720
P(satisfies | $\leq 50\,\mathrm{meV}$)	0.996	0.994	0.944	0.923	0.798
P(satisfies | $> 50\,\mathrm{meV}$)	0.987	0.979	0.912	0.884	0.642
P(satisfies | no imag. modes)	0.995	0.993	0.947	0.921	0.731
P(satisfies | imag. modes)	0.991	0.984	0.920	0.898	0.755
P(recorded | satisfies)	0.440	0.438	0.438	0.436	0.446
P(recorded | not satisfying)	0.588	0.489	0.479	0.467	0.377
not satisfying, n	119	184	1,109	1,457	3,929

Among the 3,929 structures not satisfying Set 4, one structure can violate several laws. Law 7 is violated by 2,285 structures. The stricter Law 1 threshold is violated by 487, Law 8 by 519, Law 3 by 577 and Law 6 by 404. The two site-Madelung-energy laws account for 82 violations together. Thus, violations in this population are driven mainly by the distinct-site fraction and interatomic distances.

The apparent strictness of Law 7 depends strongly on the population. Because it requires no charge assignment, its satisfaction is 71.4% across all 26,600 structures. Among the 14,284 structures with bond-valence parameters, its satisfaction is 89.7%. This subset is enriched in small, regular and well-parameterised ionic cells. Rates from a charge-free law and a charge-dependent law are therefore comparable only on a common population, as in each column above.

Two robustness checks accompany the table. First, the primary analysis evaluates Law 6 with the production CrystalNN like-charge fraction. Replacing it with the 1.15-times nearest-contact shell from an earlier sensitivity analysis changes Set 3 from 0.909 to 0.919 and Set 4 from 0.743 to 0.752.

Second, a chemistry-resolved analysis identifies the law's textbook boundary. Molecular anions include peroxides, superoxides, azides and polysulfides in which a like-charge contact is itself a covalent bond. A compositional ionicity condition cannot identify this family reliably. We therefore treat such motifs as an explicit chemical limitation rather than infer a correction from contact topology alone.

A final implementation audit found that the earlier MP-phonon table used a radial neighbour list for Law 1 and Law 8. We recomputed the reported values using the production CrystalNN definitions and tabulated bond-valence parameters. The earlier columns are retained only as implementation-sensitivity results.

Two limitations apply when relating law-set satisfaction to an experimental record. First, the sampling limitation above affects every reported rate. Second, the missing-input convention differs between the benchmark and applications. The benchmark counts structures with unavailable inputs as satisfying the law set, whereas applications report them as no verdict. Comparisons with the held-out benchmark must therefore account for both the population and the missing-input convention. Law-set and per-cell source tables accompany Supplementary Fig. S22.

S18.8 Most thresholds selected on discovery data give similar verdicts on held-out data

Every numeric threshold in the eight laws corresponds to a stated percentile of its quantity in the experimental-structure discovery distribution. We tested threshold stability by re-deriving each constant at the same percentile on the held-out split. We then measured the shift in each threshold and the fraction of verdicts that changed when the re-derived value replaced the original.

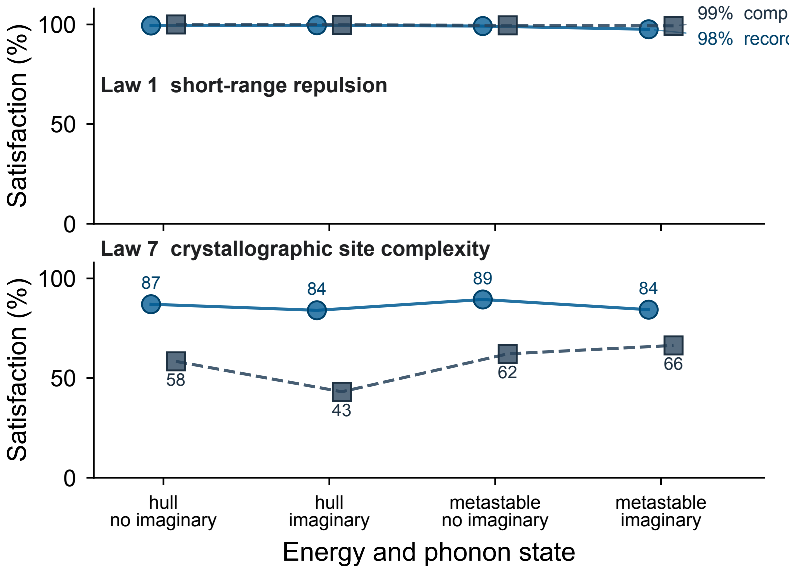

Fig. S22 PRIS satisfaction across energy, phonon and experimental-record classes. The horizontal axis crosses on-hull or metastable status with the absence or presence of imaginary phonon modes. The upper and lower profiles show satisfaction of Law 1 and Law 7, respectively. Solid blue circles denote experimentally recorded structures, and dashed grey squares denote computed-only structures. Connecting lines guide the eye.

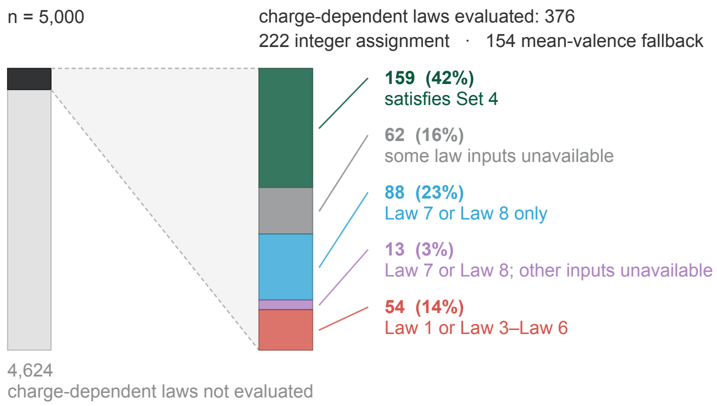

Fig. S23 Coverage of charge-dependent PRIS laws in GNoME. Flow of the random GNoME sample through formal-charge assignment and the charge-dependent laws. The evaluable branch is divided by charge-assignment route and then by Set 4 outcome and violated mechanism. Structures without a formal-charge assignment are outside the evaluable domain of the charge-dependent laws and are not classified as satisfying or failing them.

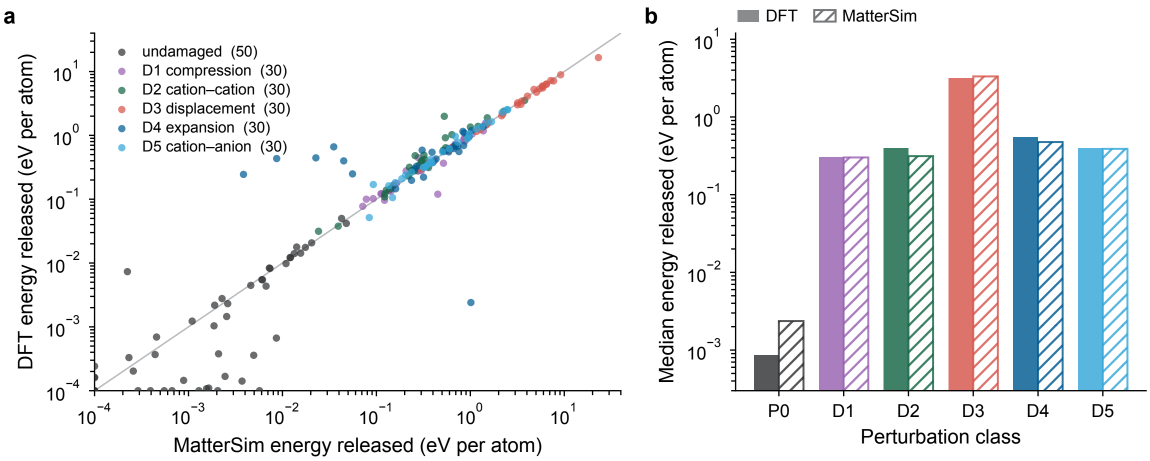

Fig. S24 The machine-learned potential against first principles on identical cells. a, Energy released on relaxation for 200 cells. The damaged cohort contains thirty discovery-split experimental parents with each of five perturbations applied. The undamaged cohort contains fifty parents, including twenty GNoME entries. DFT and MatterSim evaluated the same geometries under the same frozen-cell protocol and have a Spearman rank correlation of 0.953. Points at the axis limits denote cells for which both methods give a release below the plotted floor. **b**, The same comparison resolved by perturbation class, which exposes any class-specific disagreement. Filled bars show DFT medians, and hatched bars show MatterSim medians.

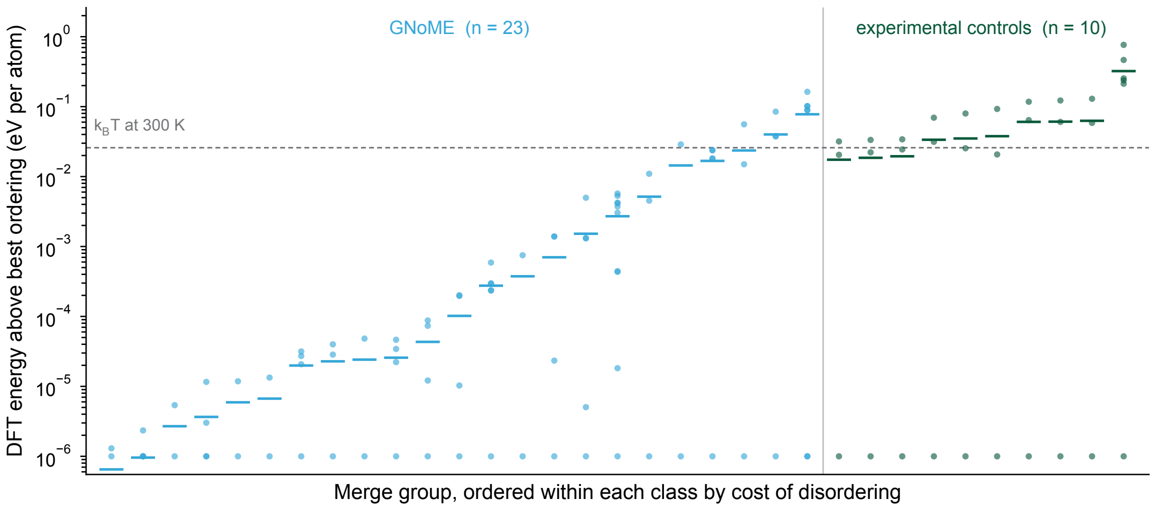

Fig. S25 Ordering energies behind the chemical-label attribution. For each merge group, every symmetry-distinct ordering of its mergeable species was relaxed with DFT. Points show each ordering's energy above the lowest-energy ordering in that group. The horizontal bar shows the cost of leaving that ordering for a random one, and groups are sorted within each class by this cost. The dashed line is $k_{\mathrm{B}}T$ at 300 K. The GNoME groups have a median disordering energy of 0.00010 eV per atom. The experimental controls have a median of 0.03642 eV per atom, which is 358 times larger. No control falls below 300 K on this measure, whereas 18 of 23 GNoME groups do. An ordering that costs less than room temperature to scramble describes a solid solution rather than an ordered crystal. Supplementary Section S20 explains which quantity distinguishes order from disorder.

threshold	original	percentile	held-out	experimental flips	damaged flips
Law 1 τ	0.735	1.0%	0.745	0.19%	1.6%
Law 1 τ (tight)	0.804	3.0%	0.809	0.17%	1.2%
Law 2 ceiling	1.05	99.4% (ionic)	1.043	0.12%	0.11%
Law 3 ceiling	1.081	96.7% (conditional)	1.065	0.64%	0.54%
Law 4 range	31.45	99.5%	26.4	0.25%	5.0%
Law 5 max	15.17	99.5%	3.9	0.33%	7.8%
Law 7 distinct sites	2/3	91.6%	3/4	2.9%	3.2%
Law 8 deviation	0.7143	96.0%	0.7143	0.00%	0.00%

The contact, packing and valence thresholds move by at most 1.5% of their values. Law 8 returns at the same grid point. The two electrostatic thresholds lie in sparsely populated tails, so their numerical values move substantially. For example, the maximum-site-energy threshold changes from 15 to 4 eV. Yet these changes flip fewer than 0.4% of experimental-structure verdicts. For deliberately permissive extreme-value laws, the verdict is therefore more stable across splits than the exact decimal threshold. Law 7 lies on a discrete grid of small-integer fractions, so its selected percentile threshold changes from 2/3 to 3/4.

S18.9 Law evaluation is negligible beside DFT relaxation

We benchmarked the implementation on one CPU core of the analysis machine. Each measurement used a rock-salt supercell and a new Python process. Law 1 requires one neighbour scan and one radius lookup. It costs 0.05 ms for an 8-atom cell, 1.0 ms for 64 atoms and 10.7 ms for 216 atoms.

The full eight-law reference implementation calls neighbour assignment three times. It costs 58 ms, 512 ms and 2.4 s on the same cells, while charge assignment adds 0.1–0.3 ms. These are measured implementation costs rather than algorithmic limits. Computing the same quantities from one cached neighbour list would be approximately three times faster.

For generative-model outputs containing 6–20 atoms per cell, Law 1 screens a queue of 10,000 structures in under one second. The unoptimised eight-law implementation takes approximately ten minutes. A literature estimate of 100 CPU hours per structure projects approximately 10^6 CPU hours for 10,000 DFT relaxations (main-text ref. 1). This is an order-of-magnitude projection, not a DFT measurement on the screened queue.

S19 Shared evaluation procedure across pre-registered analyses

Split details. The pre-registered split assigned 22,925, 9,634 and 5,748 structures to the discovery, held-out and reserve partitions, respectively. The analysis populations are the subsets with a charge assignment and all required features (Section S2). Assignment hashes structure identifiers rather than compositions. Different structures of the same composition can therefore occupy different partitions. Agreement between splits measures consistency for this random split, not performance on unseen chemistries. The composition-unseen sensitivity analysis in Section S8 restricts the frozen evaluation to reduced compositions absent from discovery, but remains a subgroup of the existing held-out partition rather than a new external cohort. The omitted-damage-type tests provide evidence beyond the damage classes used for selection.

Protocol of the fitted synthesis score. PSS (Section S15) was run under a written pre-registration fixed before fitting and archived with a time stamp and cryptographic hash. The model is an antisymmetric pairwise logistic score without an intercept. Missing features are filled with development-set medians, and all features are standardised using development-set statistics only. Pairs are weighted so that each composition contributes equally. Greedy forward selection uses grouped five-fold cross-validation, a minimum decision rate of 0.99 and a minimum required gain of 0.002, with at most eight terms.

A seeded hash split the 1,508 composition groups into 919 development groups and 589 held-out groups. A script that refuses a second run evaluated the held-out set exactly once. For full-pool transfer, the fixed coefficients and preprocessing statistics were applied without refitting. The code archive contains all coefficients, preprocessing statistics, criterion verdicts and evaluation logs.

The Set 4 analysis. The Set 4 analysis (Section S17) fixed its procedure before the augmented features existed. Set 3 was first reproduced to four decimal places on both splits, and damaged structures were recreated from the original seeds. Greedy selection maximised combined damage detection while requiring satisfaction of at least 0.81 among experimental discovery structures. Candidate thresholds came from a fixed percentile grid. The held-out split was evaluated once. Omitted-damage variants refitted only the additions above the fixed Set 3 base.

The PSS confidence breakdown in main-text Fig. 4e was examined after held-out evaluation. It is identified wherever quoted as an exploratory ordering analysis.

S20 Pre-registered first-principles tests of four learned quantities

Four quantities used in the main text were initially learned from data or machine-learning predictions rather than calculated directly with DFT. They are the contact thresholds of Law 1 and Law 2, the artificial chemical ordering behind GNoME's low-symmetry excess, the damage-severity scale predicted by a machine-learned potential, and the predicted property used to select the design queue. Each quantity was re-derived with plane-wave DFT under a protocol frozen before any calculation was submitted. The `dft/` directory of the source repository contains the pre-registration, task packages, extraction and analysis.

Protocol.

Calculations used VASP 6.3.0 with PBE PAW potentials recommended by `MPRelaxSet`. The plane-wave cutoff was $\max(520\,\mathrm{eV}, 1.3 \max \mathrm{ENMAX})$. Static calculations used an explicitly written Γ-centred mesh at $0.22\,\text{Å}^{-1}$, $\texttt{EDIFF} = 10^{-6}$ eV and tetrahedron smearing. Cell relaxations used $\texttt{ISIF} = 3$. Constant-volume points on the energy–volume curves used $\texttt{ISIF} = 4$. The campaign comprised 1,917 tasks and approximately 344 node-hours.

What each experiment measures.

Five experiments answer four questions. Two experiments map the contact landscape and provide its hard-potential control (Fig. S12a,b). The remaining three measure ordering energies (Fig. S25), test machine-learning predictions of damage severity (Fig. S24) and evaluate the property screen (Fig. S20). The ordering and property tests require distinct evaluation quantities, defined below.

For the ordering analysis, the energy above the best ordering tests whether the release selected the ground state. It does not determine whether the compound remains ordered. Thirteen of the 23 GNoME entries lie exactly at the minimum, which supports the released ground-state assignment. Order versus disorder is instead decided by the cost of leaving the best ordering for a random one. On this measure, no experimental control falls below 300 K, whereas 18 of 23 GNoME groups do. Figure S25 shows both quantities.

For the property screen, the 400 GPa target was set using machine-learning predictions of the bulk modulus. The DFT value is a median 0.940 times the predicted value, showing that the predictions run high. An unscaled absolute threshold would therefore test model calibration rather than candidate selection. Accordingly, only 0.7% of the candidates predicted to have high bulk modulus exceed an unscaled 400 GPa under DFT. The correlation between predicted and DFT values is $r = 0.769$. Thus, ranking rather than absolute scale transfers between the model and DFT, and the selection holds at the rescaled threshold (Fig. S20).

The relaxed design candidates, and what relaxing them changes.

The 260 candidates whose bulk moduli were computed are provided as Supplementary Data, with one CIF per candidate. The index reports composition, PSS and the computed modulus. It also reports space group and distinct-site fraction before and after relaxation.

Conditioning the generator on a bulk modulus above 400 GPa concentrates its output in a narrow region of the periodic table. Only eleven elements appear, and Os, Ir and Re account for almost every candidate (Fig. S21a). Relaxation also changes the symmetry used for Law 7 assessment. On the generated coordinates, 61 of the 260 candidates satisfy the Law 7 threshold. After relaxation, 113 satisfy it, and none moves in the opposite direction. This one-way change occurs because relaxation merges sites but does not split them.

The shift is concentrated among candidates that the screen retained. After relaxation, 57.9% of the high-property candidates satisfy Law 7, compared with 11.7% of the removed candidates. PSS calculated from unrelaxed coordinates therefore anticipates which cells subsequently relax into law-satisfying structures. Because relaxation data were absent from PSS construction, this provides an independent check of the screen.

Sensitivity of the fitted moduli.

The Birch–Murnaghan fit has four parameters, so a five-point energy–volume curve leaves one degree of freedom. We tested sensitivity by dropping one point from every candidate with five converged values. The fitted modulus shifts by a median of 0.05 GPa. This shift is three orders of magnitude smaller than the difference between the median moduli of retained and screened candidates.

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